

Exploring the effects of first-year cold dormancy on telomere length and mtDNA density in head-started Gopher tortoises

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Loading all the libraries that will be used in this code

```
library(dplyr) ## Will be used for specifying logical operators
library(stringr) ## Will be used to handle character vectors when modifying text data
library(tidyr) ## Will be used for reshaping the data for easier visualization later on
library(readr) ## Will be used when reading in .csv data
library(lme4) ## Will be used for mixed effect models with random and fixed effects
library(emmeans) ## Will be used for post-hoc comparisons
library(ggplot2) ## Will be used for making graphs/data visualization
```

qPCR Data Preparation

#Process EDIT all the runs for EACH qPCR Plate of samples sepearately. Use the ... Quantification Cq Results.csv files. Copy and paste the block of code and repeat on each plate of samples. You can use find replace for the dataset name, Plate1 for Plate2, etc.

Plate 1

```
# Import .csv files for each run for a particular plate of samples
Plate1_MPX <- read.csv("./Data/GT_Plate1_Multiplex_12_11_2024_Quantification_Cq_Results.csv")
dim (Plate1_MPX)
```

```
## [1] 188 16
```

```
Plate1_Telo <- read.csv("./Data/GT_Plate1_Telomeres_12_11_2024_Quantification_Cq_Results.csv")
dim (Plate1_Telo)
```

```
## [1] 94 16
```

```
# Concatenate data across runs for the same plate of samples
Plate1<- rbind(Plate1_MPX, Plate1_Telo)
dim(Plate1)
```

```
## [1] 282 16
```

```
# Add a column called PlateID and fill in correct Plate number to use as a variable in the statistics
Plate1$PlateID <- "Plate1"

# CHECK AND EDIT FOR YOUR DATA.
#Name Correct Targets based on the fluorophores used in your reaction.
Plate1$Target[Plate1$Fluor == "VIC"] <- "scnag"
Plate1$Target[Plate1$Fluor == "FAM"] <- "mtdna"
Plate1$Target[Plate1$Fluor == "SYBR"] <- "telomeres"

##### Look for outliers - 1st round #####

# Calculate the absolute difference from the mean Cq
Plate1$Diff_AVG_Cq <- abs(Plate1$Cq - Plate1$Cq.Mean)

# Identify outliers for mtdna and scnag samples based on >0.4 threshold
Plate1$Flag_outlier <- ifelse(Plate1$Target %in% c("mtdna", "scnag") &
Plate1$Diff_AVG_Cq > 0.4, "yes", "no")

# Identify outliers for telomeres based on >0.4 threshold
Plate1$Flag_outlier <- ifelse(Plate1$Target == "telomeres" &
Plate1$Diff_AVG_Cq > 0.4, "yes", Plate1$Flag_outlier)

# Report and examine high Cq samples in the first round
HighCq <- Plate1[Plate1$Flag_outlier == "yes", c("Well", "Sample", "Fluor",
"Diff_AVG_Cq")]
print(paste("Number of rows with high Cq in the first round:", nrow(HighCq)))
```

```
## [1] "Number of rows with high Cq in the first round: 8"
```

```
print(HighCq)
```

```
##      Well Sample Fluor Diff_AVG_Cq
## 51    E03 12.2_A  FAM    0.5306096
## 52    E04 12.2_A  FAM    0.5306096
## 63    F03 15.5_A  FAM    0.5237126
## 64    F04 15.5_A  FAM    0.5237126
## NA    <NA> <NA> <NA>          NA
## NA.1  <NA> <NA> <NA>          NA
## 177   H01  NEG   VIC    0.9500715
## 178   H02  NEG   VIC    0.9500714
```

```
# Remove outliers identified in the first round
Plate1 <- Plate1[Plate1$Flag_outlier != "yes", ]

# Verify the dimensions after removing outliers in the first round
print(dim(Plate1))
```

```
## [1] 276 19
```

```
##### Look for outliers - 2nd round #####

# Recalculate Cq Mean and Starting Quantity Mean based on remaining data
Plate1 <- Plate1 %>%
  group_by(Sample, Target) %>%
  mutate(
    Cq.Mean2 = mean(Cq),
    Sq.Mean2 = mean(Starting.Quantity..SQ.)
  ) %>%
  ungroup()

# Calculate the new difference from the updated mean Cq
Plate1$Diff_AVG_Cq_2 <- abs(Plate1$Cq - Plate1$Cq.Mean2)

# Identify outliers in the second round based on >0.4 threshold for telomeres
Plate1$Flag_outlier_2 <- ifelse(Plate1$Target == "telomeres" & Plate1$Diff_AVG_Cq_2 >
0.4, "yes", "no")

# Report and examine high Cq samples in the second round for telomeres
HighCq_2 <- Plate1[Plate1$Target == "telomeres" & Plate1$Diff_AVG_Cq_2 > 0.4,
c("Well", "Sample", "Fluor", "Diff_AVG_Cq_2")]
print(paste("Number of rows with high Cq in the second round for telomeres:",
nrow(HighCq_2)))
```

```
## [1] "Number of rows with high Cq in the second round for telomeres: 2"
```

```
print(HighCq_2)
```

```
## # A tibble: 2 x 4
##   Well Sample Fluor Diff_AVG_Cq_2
##   <chr> <chr> <chr>         <dbl>
## 1 <NA> <NA> <NA>           NA
## 2 <NA> <NA> <NA>           NA
```

```
# Remove outliers identified in the second round for telomeres
Plate1 <- Plate1[!(Plate1$Target == "telomeres" & Plate1$Flag_outlier_2 == "yes"), ]
```

```
# Verify the dimensions after removing outliers in the second round
print(dim(Plate1))
```

```
## [1] 276 23
```

```
### Remove samples that do not have at least two rows
```

```
Plate1 <- Plate1 %>%
  group_by(Sample) %>%
  filter(n() >= 2) %>%
  ungroup()
```

```
# Final dimensions after all filtering steps
print(dim(Plate1))
```

```
## [1] 276 23
```

```
##### Remove negative controls and standards
# rows that have "NEG", "POS" in column "Sample" and remove rows with "STD" in Sample "Content"
Plate1 <- Plate1 %>%
  filter(!str_detect(Sample, "STD")) %>%
  filter(!str_detect(Sample, "NEG")) %>%
  filter(!str_detect(Sample, "POS"))
dim(Plate1)
```

```
## [1] 234 23
```

```
##### Subset dataset based on the value in "Target" column
unique_targets <- unique(Plate1$Target)
```

```
# Create a list to store the subset dataframes
subset_dfs <- list()
```

```
# Loop through each unique value in 'Target', subset the dataframe, and store in subset_dfs
for (target_value in unique_targets) {
  subset_df <- subset(Plate1, Target == target_value)
  subset_dfs[[target_value]] <- subset_df
}
```

```
##### Now subset_dfs is a list where each element is a dataframe containing rows for each unique 'Tar
Plate1_SCNAG<-print(subset_dfs[["scnag"]])
```

```
## # A tibble: 80 x 23
##   Well Fluor X Target Content Sample Biological.Set.Name Cq Cq.Mean
##   <chr> <chr> <lg1> <chr> <chr> <chr> <lg1> <dbl> <dbl>
## 1 A03 VIC NA scnag Unkn-01 11.3_A NA 28.0 27.9
## 2 A04 VIC NA scnag Unkn-01 11.3_A NA 27.9 27.9
## 3 A05 VIC NA scnag Unkn-09 9.1_D NA 29.1 29.1
## 4 A06 VIC NA scnag Unkn-09 9.1_D NA 29.1 29.1
## 5 A07 VIC NA scnag Unkn-17 5.1_C NA 28.8 29.0
## 6 A08 VIC NA scnag Unkn-17 5.1_C NA 29.1 29.0
## 7 A09 VIC NA scnag Unkn-25 7.5_C NA 28.9 28.9
## 8 A10 VIC NA scnag Unkn-25 7.5_C NA 28.9 28.9
## 9 A11 VIC NA scnag Unkn-33 9.3_A NA 28.1 28.1
## 10 A12 VIC NA scnag Unkn-33 9.3_A NA 28.0 28.1
## # i 70 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## # Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## # Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## # Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## # Flag_outlier_2 <chr>
```

```
Plate1_SCNAG<-Plate1_SCNAG[ ,c("PlateID", "Well", "Sample", "Target", "Cq", "Cq.Mean",
"Flag_outlier", "SQ.Mean")]
```

```
Plate1_SCNAG <- Plate1_SCNAG %>%
  rename(Target_SCNAG = Target, Cq_SCNAG = Cq, Cq.Mean_SCNAG = Cq.Mean,
  Flag_outlier_SCNAG=Flag_outlier, SQ.Mean_SCNAG=SQ.Mean)
```

```
Plate1_mtDNA<-print(subset_dfs[["mtdna"]])
```

```
## # A tibble: 76 x 23
##   Well Fluor X Target Content Sample Biological.Set.Name Cq Cq.Mean
##   <chr> <chr> <lg1> <chr> <chr> <chr> <lg1> <dbl> <dbl>
## 1 A03 FAM NA mtdna Unkn-01 11.3_A NA 22.5 22.5
## 2 A04 FAM NA mtdna Unkn-01 11.3_A NA 22.5 22.5
## 3 A05 FAM NA mtdna Unkn-09 9.1_D NA 23.1 23.1
## 4 A06 FAM NA mtdna Unkn-09 9.1_D NA 23.1 23.1
## 5 A07 FAM NA mtdna Unkn-17 5.1_C NA 22.9 23.1
## 6 A08 FAM NA mtdna Unkn-17 5.1_C NA 23.4 23.1
## 7 A09 FAM NA mtdna Unkn-25 7.5_C NA 22.7 22.7
## 8 A10 FAM NA mtdna Unkn-25 7.5_C NA 22.8 22.7
## 9 A11 FAM NA mtdna Unkn-33 9.3_A NA 22.1 22.2
## 10 A12 FAM NA mtdna Unkn-33 9.3_A NA 22.2 22.2
## # i 66 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## # Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## # Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## # Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## # Flag_outlier_2 <chr>
```

```
Plate1_mtdna<-Plate1_mtdna[,c("PlateID", "Well", "Sample", "Target", "Cq",
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate1_mtdna <- Plate1_mtdna %>%
  rename(Target_mtdna = Target, Cq_mtdna = Cq, Cq.Mean_mtdna = Cq.Mean,
Flag_outlier_mtdna = Flag_outlier, SQ.Mean_mtdna = SQ.Mean)

Plate1_Telomeres<-print(subset_dfs[["telomeres"]])
```

```
## # A tibble: 78 x 23
##   Well Fluor X Target Content Sample Biological.Set.Name Cq Cq.Mean
##   <chr> <chr> <lg1> <chr> <chr> <chr> <lg1> <dbl> <dbl>
## 1 A03 SYBR NA telomeres Unkn-01 11.3_A NA 18.2 18.0
## 2 A04 SYBR NA telomeres Unkn-01 11.3_A NA 17.9 18.0
## 3 A05 SYBR NA telomeres Unkn-09 9.1_D NA 18.1 18.0
## 4 A06 SYBR NA telomeres Unkn-09 9.1_D NA 18.0 18.0
## 5 A07 SYBR NA telomeres Unkn-17 5.1_C NA 18.8 18.8
## 6 A08 SYBR NA telomeres Unkn-17 5.1_C NA 18.7 18.8
## 7 A09 SYBR NA telomeres Unkn-25 7.5_C NA 18.6 18.5
## 8 A10 SYBR NA telomeres Unkn-25 7.5_C NA 18.4 18.5
## 9 A11 SYBR NA telomeres Unkn-33 9.3_A NA 18.1 18.1
## 10 A12 SYBR NA telomeres Unkn-33 9.3_A NA 18.1 18.1
## # i 68 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## # Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## # Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## # Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## # Flag_outlier_2 <chr>
```

```
Plate1_Telomeres<-Plate1_Telomeres[,c("PlateID", "Well", "Sample", "Target",
"Cq", "Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate1_Telomeres <- Plate1_Telomeres %>%
  rename(Target_Telomeres = Target, Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean,
Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)
```

```

#rename(Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean, Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)

##### Make a final MPX dataset for by merging the Target datasets horizontally, in rows.
Plate1_FinalMPX <- merge(Plate1_SCNAG, Plate1_mtdNA, by = c("PlateID", "Well", "Sample"))

##### Normalize mtdNA
# Add a column called mtdNA, and calculate the normalized value
Plate1_FinalMPX$mtdNA <- (Plate1_FinalMPX$SQ.Mean_mtdNA / Plate1_FinalMPX$SQ.Mean_SCNAG)

# Recalculate mean across the replicates
Plate1_FinalMPX <- Plate1_FinalMPX %>%
  group_by(Sample) %>%
  mutate(
    mtdNA.Mean = mean(mtdNA)) %>%
  ungroup()

##### Merge final MPX with Telomeres
## Reduce datasets to single row per individual containing only the columns we want.
Plate1_FinalMPX <- distinct(Plate1_FinalMPX, PlateID, Sample, SQ.Mean_SCNAG,
  Cq.Mean_SCNAG, SQ.Mean_mtdNA, mtdNA.Mean)
Plate1_FinalTelo <- distinct(Plate1_Telomeres, PlateID, Sample, SQ.Mean_Telomeres,
  Cq.Mean_Telomeres)

## Merge the files horizontally
Plate1_FinalData <- merge(Plate1_FinalMPX, Plate1_FinalTelo, by = c("PlateID", "Sample"))

# Normalize Telomeres
Plate1_FinalData <- Plate1_FinalData %>% mutate(Telomeres.per.cell = SQ.Mean_Telomeres /
  SQ.Mean_SCNAG)

# Assuming Plate1_FinalData has been prepared as per your previous steps

# Aggregate to get one row per sample, taking mean of normalized mtdNA and telomeres
Plate1_FinalData <- Plate1_FinalData %>%
  group_by(Sample) %>%
  summarize(
    SQ.Mean_SCNAG = mean(SQ.Mean_SCNAG),
    SQ.Mean_mtdNA = mean(SQ.Mean_mtdNA),
    mtdNA.Mean = mean(mtdNA.Mean),
    Cq.Mean_SCNAG = mean(Cq.Mean_SCNAG),
    SQ.Mean_Telomeres = mean(SQ.Mean_Telomeres),
    Cq.Mean_Telomeres = mean(Cq.Mean_Telomeres),
    Telomeres.per.cell = mean(Telomeres.per.cell)
  ) %>%
  ungroup()

# Write the final data file for this plate
write.csv(file = "./Data/Plate1_FinalData.csv", Plate1_FinalData, row.names = FALSE)

```

Plate 2

```
# Clear memory
rm(list=ls(all = TRUE))

# Import .csv files for each run for a particular plate of samples
Plate2_MPX <- read.csv("./Data/GT_Plate2_Multiplex_12_12_2024_Quantification_Cq_Results.csv")
dim (Plate2_MPX)

## [1] 188 16

Plate2_Telo <- read.csv("./Data/GT_Plate2_Telomeres_12_12_2024_Quantification_Cq_Results.csv")
dim (Plate2_Telo)

## [1] 88 16

# Concatenate data across runs for the same plate of samples
Plate2<- rbind(Plate2_MPX, Plate2_Telo)
dim(Plate2)

## [1] 276 16

# Add a column called PlateID and fill in correct Plate number to use as a variable in the statistics
Plate2$PlateID <- "Plate2"

# CHECK AND EDIT FOR YOUR DATA.
#Name Correct Targets based on the fluorophores used in your reaction.
Plate2$Target[Plate2$Fluor == "VIC"] <- "scnag"
Plate2$Target[Plate2$Fluor == "FAM"] <- "mtdna"
Plate2$Target[Plate2$Fluor == "SYBR"] <- "telomeres"

##### Look for outliers - 1st round #####

# Calculate the absolute difference from the mean Cq
Plate2$Diff_AVG_Cq <- abs(Plate2$Cq - Plate2$Cq.Mean)

# Identify outliers for mtdna and scnag samples based on >0.4 threshold
Plate2$Flag_outlier <- ifelse(Plate2$Target %in% c("mtdna", "scnag") &
Plate2$Diff_AVG_Cq > 0.4, "yes", "no")

# Identify outliers for telomeres based on >0.4 threshold
Plate2$Flag_outlier <- ifelse(Plate2$Target == "telomeres" &
Plate2$Diff_AVG_Cq > 0.4, "yes", Plate2$Flag_outlier)

# Report and examine high Cq samples in the first round
HighCq <- Plate2[Plate2$Flag_outlier == "yes", c("Well", "Sample", "Fluor", "Diff_AVG_Cq")]
print(paste("Number of rows with high Cq in the first round:", nrow(HighCq)))

## [1] "Number of rows with high Cq in the first round: 6"
```

```
print(HighCq)
```

```
##      Well Sample Fluor Diff_AVG_Cq
## 41    D05 12.2_D   FAM   0.5404402
## 42    D06 12.2_D   FAM   0.5404402
## NA    <NA>  <NA>  <NA>         NA
## NA.1  <NA>  <NA>  <NA>         NA
## NA.2  <NA>  <NA>  <NA>         NA
## NA.3  <NA>  <NA>  <NA>         NA
```

```
# Remove outliers identified in the first round
Plate2 <- Plate2[Plate2$Flag_outlier != "yes", ]
```

```
# Verify the dimensions after removing outliers in the first round
print(dim(Plate2))
```

```
## [1] 274 19
```

```
##### Look for outliers - 2nd round #####
```

```
# Recalculate Cq Mean and Starting Quantity Mean based on remaining data
```

```
Plate2 <- Plate2 %>%
  group_by(Sample, Target) %>%
  mutate(
    Cq.Mean2 = mean(Cq),
    Sq.Mean2 = mean(Starting.Quantity..SQ.)
  ) %>%
  ungroup()
```

```
# Calculate the new difference from the updated mean Cq
Plate2$Diff_AVG_Cq_2 <- abs(Plate2$Cq - Plate2$Cq.Mean2)
```

```
# Identify outliers in the second round based on >0.4 threshold for telomeres
Plate2$Flag_outlier_2 <- ifelse(Plate2$Target == "telomeres" & Plate2$Diff_AVG_Cq_2 >
0.4, "yes", "no")
```

```
# Report and examine high Cq samples in the second round for telomeres
HighCq_2 <- Plate2[Plate2$Target == "telomeres" & Plate2$Diff_AVG_Cq_2 > 0.4,
c("Well", "Sample", "Fluor", "Diff_AVG_Cq_2")]
print(paste("Number of rows with high Cq in the second round for telomeres:",
nrow(HighCq_2)))
```

```
## [1] "Number of rows with high Cq in the second round for telomeres: 4"
```

```
print(HighCq_2)
```

```
## # A tibble: 4 x 4
##   Well Sample Fluor Diff_AVG_Cq_2
##   <chr> <chr>  <chr>         <dbl>
## 1 <NA>  <NA>    <NA>         NA
## 2 <NA>  <NA>    <NA>         NA
## 3 <NA>  <NA>    <NA>         NA
## 4 <NA>  <NA>    <NA>         NA
```



```
# Remove outliers identified in the second round for telomeres
Plate2 <- Plate2[!(Plate2$Target == "telomeres" & Plate2$Flag_outlier_2 == "yes"), ]

# Verify the dimensions after removing outliers in the second round
print(dim(Plate2))
```

```
## [1] 274 23
```

```
### Remove samples that do not have at least two rows
Plate2 <- Plate2 %>%
  group_by(Sample) %>%
  filter(n() >= 2) %>%
  ungroup()
```

```
# Final dimensions after all filtering steps
print(dim(Plate2))
```

```
## [1] 274 23
```

```
##### Remove negative controls and standards
# rows that have "NEG", "POS" in column "Sample" and remove rows with "STD" in Sample "Content"
Plate2 <- Plate2 %>%
  filter(!str_detect(Sample, "STD")) %>%
  filter(!str_detect(Sample, "NEG"))
dim(Plate2)
```

```
## [1] 236 23
```

```
##### Subset dataset based on the value in "Target" column
unique_targets <- unique(Plate2$Target)
```

```
# Create a list to store the subset dataframes
subset_dfs <- list()
```

```
# Loop through each unique value in 'Target', subset the dataframe, and store in subset_dfs
```

```
for (target_value in unique_targets) {
  subset_df <- subset(Plate2, Target == target_value)
  subset_dfs[[target_value]] <- subset_df
}
```

```
#####
```

```
#####Now subset_dfs is a list where each element is a dataframe containing rows for each unique
Plate2_SCNAG<-print(subset_dfs[["scnag"]])
```

```
## # A tibble: 80 x 23
```

```
##   X      Well Fluor Target Content Sample Biological.Set.Name    Cq Cq.Mean
##   <lg1> <chr> <chr> <chr> <chr> <chr> <lg1> <dbl> <dbl>
## 1 NA    A03   VIC   scnag  Unkn-01 15.3_D NA      29.5   29.3
## 2 NA    A04   VIC   scnag  Unkn-01 15.3_D NA      29.1   29.3
## 3 NA    A05   VIC   scnag  Unkn-09 5.5_A  NA      28.8   28.8
```

```
## 4 NA A06 VIC scnag Unkn-09 5.5_A NA 28.8 28.8
## 5 NA A07 VIC scnag Unkn-17 6.3_D NA 29.7 29.8
## 6 NA A08 VIC scnag Unkn-17 6.3_D NA 29.9 29.8
## 7 NA A09 VIC scnag Unkn-25 10.1_C NA 29.9 29.8
## 8 NA A10 VIC scnag Unkn-25 10.1_C NA 29.7 29.8
## 9 NA A11 VIC scnag Unkn-33 15.1_C NA 28.2 28.2
## 10 NA A12 VIC scnag Unkn-33 15.1_C NA 28.2 28.2
## # i 70 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## # Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## # Set.Point <int>, Well.Note <lgl>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## # Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## # Flag_outlier_2 <chr>
```

```
Plate2_SCNAG<-Plate2_SCNAG[,c("PlateID", "Well", "Sample", "Target", "Cq", "Cq.Mean",
"Flag_outlier", "SQ.Mean")]
Plate2_SCNAG <- Plate2_SCNAG %>%
  rename(Target_SCNAG = Target, Cq_SCNAG = Cq, Cq.Mean_SCNAG = Cq.Mean,
  Flag_outlier_SCNAG=Flag_outlier, SQ.Mean_SCNAG=SQ.Mean)

Plate2_mtDNA<-print(subset_dfs[["mtdna"]])
```

```
## # A tibble: 78 x 23
## X Well Fluor Target Content Sample Biological.Set.Name Cq Cq.Mean
## <lgl> <chr> <chr> <chr> <chr> <chr> <lgl> <dbl> <dbl>
## 1 NA A03 FAM mtdna Unkn-01 15.3_D NA 22.7 22.5
## 2 NA A04 FAM mtdna Unkn-01 15.3_D NA 22.4 22.5
## 3 NA A05 FAM mtdna Unkn-09 5.5_A NA 22.8 22.8
## 4 NA A06 FAM mtdna Unkn-09 5.5_A NA 22.8 22.8
## 5 NA A07 FAM mtdna Unkn-17 6.3_D NA 23.2 23.3
## 6 NA A08 FAM mtdna Unkn-17 6.3_D NA 23.3 23.3
## 7 NA A09 FAM mtdna Unkn-25 10.1_C NA 23.1 23.1
## 8 NA A10 FAM mtdna Unkn-25 10.1_C NA 23.1 23.1
## 9 NA A11 FAM mtdna Unkn-33 15.1_C NA 22.2 22.2
## 10 NA A12 FAM mtdna Unkn-33 15.1_C NA 22.1 22.2
## # i 68 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## # Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## # Set.Point <int>, Well.Note <lgl>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## # Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## # Flag_outlier_2 <chr>
```

```
Plate2_mtDNA<-Plate2_mtDNA[,c("PlateID", "Well", "Sample", "Target", "Cq",
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate2_mtDNA <- Plate2_mtDNA %>%
  rename(Target_mtDNA = Target, Cq_mtDNA = Cq, Cq.Mean_mtDNA = Cq.Mean,
  Flag_outlier_mtDNA = Flag_outlier, SQ.Mean_mtDNA = SQ.Mean)

Plate2_Telomeres<-print(subset_dfs[["telomeres"]])
```

```
## # A tibble: 78 x 23
## X Well Fluor Target Content Sample Biological.Set.Name Cq Cq.Mean
## <lgl> <chr> <chr> <chr> <chr> <chr> <lgl> <dbl> <dbl>
```

```
## 1 NA A03 SYBR telomeres Unkn-01 15.3_D NA 18.4 18.4
## 2 NA A04 SYBR telomeres Unkn-01 15.3_D NA 18.4 18.4
## 3 NA A05 SYBR telomeres Unkn-09 5.5_A NA 18.3 18.4
## 4 NA A06 SYBR telomeres Unkn-09 5.5_A NA 18.4 18.4
## 5 NA A07 SYBR telomeres Unkn-17 6.3_D NA 18.7 18.8
## 6 NA A08 SYBR telomeres Unkn-17 6.3_D NA 19.0 18.8
## 7 NA A09 SYBR telomeres Unkn-25 10.1_C NA 18.2 18.3
## 8 NA A10 SYBR telomeres Unkn-25 10.1_C NA 18.3 18.3
## 9 NA A11 SYBR telomeres Unkn-33 15.1_C NA 18.2 18.3
## 10 NA A12 SYBR telomeres Unkn-33 15.1_C NA 18.4 18.3
## # i 68 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## # Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## # Set.Point <int>, Well.Note <lgl>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## # Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## # Flag_outlier_2 <chr>
```

```
Plate2_Telomeres<-Plate2_Telomeres[ ,c("PlateID", "Well", "Sample", "Target",
"Cq", "Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate2_Telomeres <- Plate2_Telomeres %>%
  rename(Target_Telomeres = Target, Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean,
  Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)
  #rename(Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean, Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)

#####. Make a final MPX dataset for by merging the Target datasets horizontally, in rows.
Plate2_FinalMPX <- merge(Plate2_SCNAG, Plate2_mtDNA, by = c("PlateID", "Well", "Sample"))

##### Normalize mtDNA
# Add a column called mtDNA, and calculate the normalized value
Plate2_FinalMPX$mtDNA <- (Plate2_FinalMPX$SQ.Mean_mtDNA / Plate2_FinalMPX$SQ.Mean_SCNAG)

# Recalculate mean across the replicates
Plate2_FinalMPX <- Plate2_FinalMPX %>%
  group_by(Sample) %>%
  mutate(
    mtDNA.Mean = mean(mtDNA)) %>%
  ungroup()

##### Merge final MPX with Telomeres
## Reduce datasets to single row per individual containing only the columns we want.
Plate2_FinalMPX <- distinct(Plate2_FinalMPX, PlateID, Sample, SQ.Mean_SCNAG,
Cq.Mean_SCNAG, SQ.Mean_mtDNA, mtDNA.Mean)
Plate2_FinalTelo <- distinct(Plate2_Telomeres, PlateID, Sample, SQ.Mean_Telomeres,
Cq.Mean_Telomeres)

## Merge the files horizontally
Plate2_FinalData <- merge(Plate2_FinalMPX, Plate2_FinalTelo, by = c("PlateID", "Sample"))

# Normalize Telomeres
Plate2_FinalData <- Plate2_FinalData %>% mutate(Telomeres.per.cell = SQ.Mean_Telomeres /
SQ.Mean_SCNAG)

# Assuming Plate2_FinalData has been prepared as per your previous steps
```

```

# Aggregate to get one row per sample, taking mean of normalized mtDNA and telomeres
Plate2_FinalData <- Plate2_FinalData %>%
  group_by(Sample) %>%
  summarize(
    SQ.Mean_SCNAG = mean(SQ.Mean_SCNAG),
    SQ.Mean_mtDNA = mean(SQ.Mean_mtDNA),
    mtDNA.Mean = mean(mtDNA.Mean),
    Cq.Mean_SCNAG = mean(Cq.Mean_SCNAG),
    SQ.Mean_Telomeres = mean(SQ.Mean_Telomeres),
    Cq.Mean_Telomeres = mean(Cq.Mean_Telomeres),
    Telomeres.per.cell = mean(Telomeres.per.cell)
  ) %>%
  ungroup()

# Write the final data file for this plate
write.csv(file = "./Data/Plate2_FinalData.csv", Plate2_FinalData, row.names = FALSE)

```

Plate 3

```

# Clear memory
rm(list=ls(all = TRUE))

# Import .csv files for each run for a particular plate of samples
Plate3_MPX <- read.csv("./Data/GT_Plate3_Multiplex_12_16_2024_Quantification_Cq_Results.csv")
dim (Plate3_MPX)

```

```
## [1] 180 16
```

```

Plate3_Telo <- read.csv("./Data/GT_Plate3_Telomeres_12_16_2024_Quantification_Cq_Results.csv")
dim (Plate3_Telo)

```

```
## [1] 86 16
```

```

# Concatenate data across runs for the same plate of samples
Plate3<- rbind(Plate3_MPX, Plate3_Telo)
dim(Plate3)

```

```
## [1] 266 16
```

```

# Add a column called PlateID and fill in correct Plate number to use as a variable in the statistics
Plate3$PlateID <- "Plate3"

```

```

# CHECK AND EDIT FOR YOUR DATA.
#Name Correct Targets based on the fluorophores used in your reaction.
Plate3$Target[Plate3$Fluor == "VIC"] <- "scnag"
Plate3$Target[Plate3$Fluor == "FAM"] <- "mtdna"
Plate3$Target[Plate3$Fluor == "SYBR"] <- "telomeres"

```

```
##### Look for outliers - 1st round #####
```

```

# Calculate the absolute difference from the mean Cq
Plate3$Diff_AVG_Cq <- abs(Plate3$Cq - Plate3$Cq.Mean)

# Identify outliers for mtdna and scnag samples based on >0.4 threshold
Plate3$Flag_outlier <- ifelse(Plate3$Target %in% c("mtdna", "scnag") & Plate3$Diff_AVG_Cq >
0.4, "yes", "no")

# Identify outliers for telomeres based on >0.4 threshold
Plate3$Flag_outlier <- ifelse(Plate3$Target == "telomeres" & Plate3$Diff_AVG_Cq > 0.4,
"yes", Plate3$Flag_outlier)

# Report and examine high Cq samples in the first round
HighCq <- Plate3[Plate3$Flag_outlier == "yes", c("Well", "Sample", "Fluor", "Diff_AVG_Cq")]
print(paste("Number of rows with high Cq in the first round:", nrow(HighCq)))

```

```
## [1] "Number of rows with high Cq in the first round: 4"
```

```
print(HighCq)
```

```
##      Well Sample Fluor Diff_AVG_Cq
## NA      <NA>      <NA>      <NA>      NA
## NA.1 <NA>      <NA>      <NA>      NA
## 169   H01      NEG      VIC      0.7245049
## 170   H02      NEG      VIC      0.7245049
```

```

# Remove outliers identified in the first round
Plate3 <- Plate3[Plate3$Flag_outlier != "yes", ]

# Verify the dimensions after removing outliers in the first round
print(dim(Plate3))

```

```
## [1] 264 19
```

```
##### Look for outliers - 2nd round #####
```

```

# Recalculate Cq Mean and Starting Quantity Mean based on remaining data
Plate3 <- Plate3 %>%
  group_by(Sample, Target) %>%
  mutate(
    Cq.Mean2 = mean(Cq),
    Sq.Mean2 = mean(Starting.Quantity..SQ.)
  ) %>%
  ungroup()

```

```

# Calculate the new difference from the updated mean Cq
Plate3$Diff_AVG_Cq_2 <- abs(Plate3$Cq - Plate3$Cq.Mean2)

# Identify outliers in the second round based on >0.4 threshold for telomeres
Plate3$Flag_outlier_2 <- ifelse(Plate3$Target == "telomeres" & Plate3$Diff_AVG_Cq_2 >
0.4, "yes", "no")

```

```
# Report and examine high Cq samples in the second round for telomeres
HighCq_2 <- Plate3[Plate3$Target == "telomeres" & Plate3$Diff_AVG_Cq_2 > 0.4,
c("Well", "Sample", "Fluor", "Diff_AVG_Cq_2")]
print(paste("Number of rows with high Cq in the second round for telomeres:", nrow(HighCq_2)))
```

```
## [1] "Number of rows with high Cq in the second round for telomeres: 2"
```

```
print(HighCq_2)
```

```
## # A tibble: 2 x 4
##   Well Sample Fluor Diff_AVG_Cq_2
##   <chr> <chr> <chr>         <dbl>
## 1 <NA> <NA> <NA>             NA
## 2 <NA> <NA> <NA>             NA
```

```
# Remove outliers identified in the second round for telomeres
Plate3 <- Plate3[!(Plate3$Target == "telomeres" & Plate3$Flag_outlier_2 == "yes"), ]
```

```
# Verify the dimensions after removing outliers in the second round
print(dim(Plate3))
```

```
## [1] 264 23
```

```
#### Remove samples that do not have at least two rows
Plate3 <- Plate3 %>%
  group_by(Sample) %>%
  filter(n() >= 2) %>%
  ungroup()
```

```
# Final dimensions after all filtering steps
print(dim(Plate3))
```

```
## [1] 264 23
```

```
##### Remove negative controls and standards
# rows that have "NEG", "POS" in column "Sample" and remove rows with "STD" in Sample "Content"
Plate3 <- Plate3 %>%
  filter(!str_detect(Sample, "STD")) %>%
  filter(!str_detect(Sample, "NEG"))
dim(Plate3)
```

```
## [1] 228 23
```

```
##### Subset dataset based on the value in "Target" column
unique_targets <- unique(Plate3$Target)
# Create a list to store the subset dataframes
subset_dfs <- list()
# Loop through each unique value in 'Target', subset the dataframe, and store in subset_dfs
for (target_value in unique_targets) {
  subset_df <- subset(Plate3, Target == target_value)
```

```
subset_dfs[[target_value]] <- subset_df
}
```

Now subset_dfs is a list where each element is a dataframe containing rows for each unique Plate3_SCNAG<-print(subset_dfs[["scnag"]])

```
## # A tibble: 76 x 23
##   X      Well Fluor Target Content Sample Biological.Set.Name    Cq Cq.Mean
##   <lg1> <chr> <chr> <chr> <chr> <chr> <lg1>          <dbl> <dbl>
## 1 NA     A03   VIC   scnag Unkn-01 7.2_D NA           27.3   27.2
## 2 NA     A04   VIC   scnag Unkn-01 7.2_D NA           27.2   27.2
## 3 NA     A05   VIC   scnag Unkn-09 2.3_D NA           27.1   27.1
## 4 NA     A06   VIC   scnag Unkn-09 2.3_D NA           27.2   27.1
## 5 NA     A07   VIC   scnag Unkn-17 5.4_C NA           28.3   28.3
## 6 NA     A08   VIC   scnag Unkn-17 5.4_C NA           28.3   28.3
## 7 NA     A09   VIC   scnag Unkn-25 8.1_B NA           26.6   26.6
## 8 NA     A10   VIC   scnag Unkn-25 8.1_B NA           26.5   26.6
## 9 NA     A11   VIC   scnag Unkn-33 2.2_A NA           27.8   27.5
## 10 NA    A12   VIC   scnag Unkn-33 2.2_A NA           27.3   27.5
## # i 66 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate3_SCNAG<-Plate3_SCNAG[ ,c("PlateID", "Well", "Sample", "Target", "Cq", "Cq.Mean",
"Flag_outlier", "SQ.Mean")]
```

```
Plate3_SCNAG <- Plate3_SCNAG %>%
  rename(Target_SCNAG = Target, Cq_SCNAG = Cq, Cq.Mean_SCNAG = Cq.Mean,
  Flag_outlier_SCNAG=Flag_outlier, SQ.Mean_SCNAG=SQ.Mean)
```

```
Plate3_mtDNA<-print(subset_dfs[["mtdna"]])
```

```
## # A tibble: 76 x 23
##   X      Well Fluor Target Content Sample Biological.Set.Name    Cq Cq.Mean
##   <lg1> <chr> <chr> <chr> <chr> <chr> <lg1>          <dbl> <dbl>
## 1 NA     A03   FAM   mtdna Unkn-01 7.2_D NA           21.5   21.5
## 2 NA     A04   FAM   mtdna Unkn-01 7.2_D NA           21.4   21.5
## 3 NA     A05   FAM   mtdna Unkn-09 2.3_D NA           22.0   22.0
## 4 NA     A06   FAM   mtdna Unkn-09 2.3_D NA           22.0   22.0
## 5 NA     A07   FAM   mtdna Unkn-17 5.4_C NA           22.7   22.7
## 6 NA     A08   FAM   mtdna Unkn-17 5.4_C NA           22.7   22.7
## 7 NA     A09   FAM   mtdna Unkn-25 8.1_B NA           20.9   20.8
## 8 NA     A10   FAM   mtdna Unkn-25 8.1_B NA           20.8   20.8
## 9 NA     A11   FAM   mtdna Unkn-33 2.2_A NA           22.0   21.6
## 10 NA    A12   FAM   mtdna Unkn-33 2.2_A NA           21.3   21.6
## # i 66 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```

Plate3_mtDNA<-Plate3_mtDNA[ ,c("PlateID", "Well", "Sample", "Target", "Cq",
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate3_mtDNA <- Plate3_mtDNA %>%
  rename(Target_mtDNA = Target, Cq_mtDNA = Cq, Cq.Mean_mtDNA = Cq.Mean,
  Flag_outlier_mtDNA = Flag_outlier, SQ.Mean_mtDNA = SQ.Mean)

Plate3_Telomeres<-print(subset_dfs[["telomeres"]])

```

```

## # A tibble: 76 x 23
##   X      Well Fluor Target   Content Sample Biological.Set.Name   Cq Cq.Mean
##   <lg1> <chr> <chr> <chr>    <chr>   <chr>   <lg1>          <dbl> <dbl>
## 1 NA    A03  SYBR telomeres Unkn-01 7.2_D NA          17.8  17.9
## 2 NA    A04  SYBR telomeres Unkn-01 7.2_D NA          18.0  17.9
## 3 NA    A05  SYBR telomeres Unkn-09 2.3_D NA          17.6  17.6
## 4 NA    A06  SYBR telomeres Unkn-09 2.3_D NA          17.6  17.6
## 5 NA    A07  SYBR telomeres Unkn-17 5.4_C NA          17.8  18.0
## 6 NA    A08  SYBR telomeres Unkn-17 5.4_C NA          18.2  18.0
## 7 NA    A09  SYBR telomeres Unkn-25 8.1_B NA          17.4  17.3
## 8 NA    A10  SYBR telomeres Unkn-25 8.1_B NA          17.3  17.3
## 9 NA    B03  SYBR telomeres Unkn-02 10.2_A NA         17.7  17.6
## 10 NA   B04  SYBR telomeres Unkn-02 10.2_A NA         17.6  17.6
## # i 66 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>

```

```

Plate3_Telomeres<-Plate3_Telomeres[ ,c("PlateID", "Well", "Sample", "Target",
"Cq", "Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate3_Telomeres <- Plate3_Telomeres %>%
  rename(Target_Telomeres = Target, Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean,
  Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)
  #rename(Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean, Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)

#####. Make a final MPX dataset for by merging the Target datasets horizontally, in rows.
Plate3_FinalMPX <- merge(Plate3_SCNAG, Plate3_mtDNA, by = c("PlateID", "Well", "Sample"))

##### Normalize mtDNA
# Add a column called mtDNA, and calculate the normalized value
Plate3_FinalMPX$mtDNA <- (Plate3_FinalMPX$SQ.Mean_mtDNA / Plate3_FinalMPX$SQ.Mean_SCNAG)

# Recalculate mean across the replicates
Plate3_FinalMPX <- Plate3_FinalMPX %>%
  group_by(Sample) %>%
  mutate(
    mtDNA.Mean = mean(mtDNA)) %>%
  ungroup()

##### Merge final MPX with Telomeres
## Reduce datasets to single row per individual containing only the columns we want.
Plate3_FinalMPX <- distinct(Plate3_FinalMPX, PlateID, Sample, SQ.Mean_SCNAG,
Cq.Mean_SCNAG, SQ.Mean_mtDNA, mtDNA.Mean)

```



```

Plate3_FinalTelo <- distinct(Plate3_Telomeres, PlateID, Sample, SQ.Mean_Telomeres,
Cq.Mean_Telomeres)

## Merge the files horizontally
Plate3_FinalData <- merge(Plate3_FinalMPX, Plate3_FinalTelo, by = c("PlateID", "Sample"))

# Normalize Telomeres
Plate3_FinalData <- Plate3_FinalData %>% mutate(Telomeres.per.cell = SQ.Mean_Telomeres /
SQ.Mean_SCNAG)

# Assuming Plate3_FinalData has been prepared as per your previous steps

# Aggregate to get one row per sample, taking mean of normalized mtDNA and telomeres
Plate3_FinalData <- Plate3_FinalData %>%
  group_by(Sample) %>%
  summarize(
    SQ.Mean_SCNAG = mean(SQ.Mean_SCNAG),
    SQ.Mean_mtDNA = mean(SQ.Mean_mtDNA),
    mtDNA.Mean = mean(mtDNA.Mean),
    Cq.Mean_SCNAG = mean(Cq.Mean_SCNAG),
    SQ.Mean_Telomeres = mean(SQ.Mean_Telomeres),
    Cq.Mean_Telomeres = mean(Cq.Mean_Telomeres),
    Telomeres.per.cell = mean(Telomeres.per.cell)
  ) %>%
  ungroup()

# Write the final data file for this plate
write.csv(file = "./Data/Plate3_FinalData.csv", Plate3_FinalData, row.names = FALSE)

```

Plate 4

```

# Clear memory
rm(list=ls(all = TRUE))

# Import .csv files for each run for a particular plate of samples
Plate4_MPX <- read.csv("./Data/GT_Plate4_Multiplex_12_17_2024_Quantification_Cq_Results.csv")
dim (Plate4_MPX)

## [1] 180 16

Plate4_Telo <- read.csv("./Data/GT_Plate4_Telomeres_12_17_2024_Quantification_Cq_Results.csv")
dim (Plate4_Telo)

## [1] 82 16

# Concatenate data across runs for the same plate of samples
Plate4<- rbind(Plate4_MPX, Plate4_Telo)
dim(Plate4)

## [1] 262 16

```

```
# Add a column called PlateID and fill in correct Plate number to use as a variable in the statistics
Plate4$PlateID <- "Plate4"
```

```
# CHECK AND EDIT FOR YOUR DATA.
```

```
#Name Correct Targets based on the fluorophores used in your reaction.
```

```
Plate4$Target[Plate4$Fluor == "VIC"] <- "scnag"
```

```
Plate4$Target[Plate4$Fluor == "FAM"] <- "mtdna"
```

```
Plate4$Target[Plate4$Fluor == "SYBR"] <- "telomeres"
```

```
##### Look for outliers - 1st round #####
```

```
# Calculate the absolute difference from the mean Cq
```

```
Plate4$Diff_AVG_Cq <- abs(Plate4$Cq - Plate4$Cq.Mean)
```

```
# Identify outliers for mtdna and scnag samples based on >0.4 threshold
```

```
Plate4$Flag_outlier <- ifelse(Plate4$Target %in% c("mtdna", "scnag") &
```

```
Plate4$Diff_AVG_Cq > 0.4, "yes", "no")
```

```
# Identify outliers for telomeres based on >0.4 threshold
```

```
Plate4$Flag_outlier <- ifelse(Plate4$Target == "telomeres" & Plate4$Diff_AVG_Cq >
0.4, "yes", Plate4$Flag_outlier)
```

```
# Report and examine high Cq samples in the first round
```

```
HighCq <- Plate4[Plate4$Flag_outlier == "yes", c("Well", "Sample", "Fluor", "Diff_AVG_Cq")]
print(paste("Number of rows with high Cq in the first round:", nrow(HighCq)))
```

```
## [1] "Number of rows with high Cq in the first round: 5"
```

```
print(HighCq)
```

```
##      Well Sample Fluor Diff_AVG_Cq
## 57    F01   STD6   FAM    0.4278492
## 58    F02   STD6   FAM    0.4278492
## NA    <NA>   <NA>   <NA>          NA
## NA.1  <NA>   <NA>   <NA>          NA
## NA.2  <NA>   <NA>   <NA>          NA
```

```
# Remove outliers identified in the first round
```

```
Plate4 <- Plate4[Plate4$Flag_outlier != "yes", ]
```

```
# Verify the dimensions after removing outliers in the first round
```

```
print(dim(Plate4))
```

```
## [1] 260 19
```

```
##### Look for outliers - 2nd round #####
```

```
# Recalculate Cq Mean and Starting Quantity Mean based on remaining data
```

```
Plate4 <- Plate4 %>%
```

```
  group_by(Sample, Target) %>%
```

```
  mutate(
```

```

    Cq.Mean2 = mean(Cq),
    Sq.Mean2 = mean(Starting.Quantity..SQ.)
  ) %>%
  ungroup()

# Calculate the new difference from the updated mean Cq
Plate4$Diff_AVG_Cq_2 <- abs(Plate4$Cq - Plate4$Cq.Mean2)

# Identify outliers in the second round based on >0.4 threshold for telomeres
Plate4$Flag_outlier_2 <- ifelse(Plate4$Target == "telomeres" & Plate4$Diff_AVG_Cq_2 >
0.4, "yes", "no")

# Report and examine high Cq samples in the second round for telomeres
HighCq_2 <- Plate4[Plate4$Target == "telomeres" & Plate4$Diff_AVG_Cq_2 > 0.4,
c("Well", "Sample", "Fluor", "Diff_AVG_Cq_2")]
print(paste("Number of rows with high Cq in the second round for telomeres:", nrow(HighCq_2)))

## [1] "Number of rows with high Cq in the second round for telomeres: 3"

print(HighCq_2)

## # A tibble: 3 x 4
##   Well Sample Fluor Diff_AVG_Cq_2
##   <chr> <chr> <chr>         <dbl>
## 1 <NA> <NA> <NA>             NA
## 2 <NA> <NA> <NA>             NA
## 3 <NA> <NA> <NA>             NA

# Remove outliers identified in the second round for telomeres
Plate4 <- Plate4[!(Plate4$Target == "telomeres" & Plate4$Flag_outlier_2 == "yes"), ]

# Verify the dimensions after removing outliers in the second round
print(dim(Plate4))

## [1] 260 23

### Remove samples that do not have at least two rows
Plate4 <- Plate4 %>%
  group_by(Sample) %>%
  filter(n() >= 2) %>%
  ungroup()

# Final dimensions after all filtering steps
print(dim(Plate4))

## [1] 260 23

##### Remove negative controls and standards
# rows that have "NEG", "POS" in column "Sample" and remove rows with "STD" in Sample "Content"
Plate4 <- Plate4 %>%
  filter(!str_detect(Sample, "STD")) %>%
  filter(!str_detect(Sample, "NEG"))
dim(Plate4)

```

```
## [1] 224 23
```

```
##### Subset dataset based on the value in "Target" column
```

```
unique_targets <- unique(Plate4$Target)
```

```
# Create a list to store the subset dataframes
```

```
subset_dfs <- list()
```

```
# Loop through each unique value in 'Target', subset the dataframe, and store in subset_dfs
```

```
for (target_value in unique_targets) {  
  subset_df <- subset(Plate4, Target == target_value)  
  subset_dfs[[target_value]] <- subset_df  
}
```

```
##### Now subset_dfs is a list where each element is a dataframe containing rows for each unique
```

```
Plate4_SCNAG<-print(subset_dfs[["scnag"]])
```

```
## # A tibble: 76 x 23
```

##	X	Well	Fluor	Target	Content	Sample	Biological.Set.Name	Cq	Cq.Mean
##	<lgl>	<chr>	<chr>	<chr>	<chr>	<chr>	<lgl>	<dbl>	<dbl>
##	1	NA	A03	VIC	scnag	Unkn-01 7.1_B	NA	27.5	27.4
##	2	NA	A04	VIC	scnag	Unkn-01 7.1_B	NA	27.4	27.4
##	3	NA	A05	VIC	scnag	Unkn-09 15.5_B	NA	28.3	28.3
##	4	NA	A06	VIC	scnag	Unkn-09 15.5_B	NA	28.3	28.3
##	5	NA	A07	VIC	scnag	Unkn-17 17.2_D	NA	27.0	27.0
##	6	NA	A08	VIC	scnag	Unkn-17 17.2_D	NA	27.0	27.0
##	7	NA	A09	VIC	scnag	Unkn-25 8.2_A	NA	28.2	27.9
##	8	NA	A10	VIC	scnag	Unkn-25 8.2_A	NA	27.6	27.9
##	9	NA	A11	VIC	scnag	Unkn-33 15.3_C	NA	27.4	27.4
##	10	NA	A12	VIC	scnag	Unkn-33 15.3_C	NA	27.5	27.4

```
## # i 66 more rows
```

```
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
```

```
## # Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
```

```
## # Set.Point <int>, Well.Note <lgl>, PlateID <chr>, Diff_AVG_Cq <dbl>,
```

```
## # Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
```

```
## # Flag_outlier_2 <chr>
```

```
Plate4_SCNAG<-Plate4_SCNAG[,c("PlateID", "Well", "Sample", "Target", "Cq",  
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
```

```
Plate4_SCNAG <- Plate4_SCNAG %>%
```

```
  rename(Target_SCNAG = Target, Cq_SCNAG = Cq, Cq.Mean_SCNAG = Cq.Mean,
```

```
  Flag_outlier_SCNAG=Flag_outlier, SQ.Mean_SCNAG=SQ.Mean)
```

```
Plate4_mtdna<-print(subset_dfs[["mtdna"]])
```

```
## # A tibble: 76 x 23
```

##	X	Well	Fluor	Target	Content	Sample	Biological.Set.Name	Cq	Cq.Mean
##	<lgl>	<chr>	<chr>	<chr>	<chr>	<chr>	<lgl>	<dbl>	<dbl>
##	1	NA	A03	FAM	mtdna	Unkn-01 7.1_B	NA	21.6	21.5
##	2	NA	A04	FAM	mtdna	Unkn-01 7.1_B	NA	21.4	21.5
##	3	NA	A05	FAM	mtdna	Unkn-09 15.5_B	NA	22.0	22.0
##	4	NA	A06	FAM	mtdna	Unkn-09 15.5_B	NA	22.0	22.0
##	5	NA	A07	FAM	mtdna	Unkn-17 17.2_D	NA	21.5	21.6
##	6	NA	A08	FAM	mtdna	Unkn-17 17.2_D	NA	21.6	21.6

```
## 7 NA      A09      FAM      mtdna  Unkn-25 8.2_A  NA      21.7      21.5
## 8 NA      A10      FAM      mtdna  Unkn-25 8.2_A  NA      21.2      21.5
## 9 NA      A11      FAM      mtdna  Unkn-33 15.3_C NA      22.0      22.0
## 10 NA     A12      FAM      mtdna  Unkn-33 15.3_C NA      22.0      22.0
## # i 66 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lgl>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate4_mtDNA<-Plate4_mtDNA[,c("PlateID", "Well", "Sample", "Target", "Cq",
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate4_mtDNA <- Plate4_mtDNA %>%
  rename(Target_mtDNA = Target, Cq_mtDNA = Cq, Cq.Mean_mtDNA = Cq.Mean,
  Flag_outlier_mtDNA = Flag_outlier, SQ.Mean_mtDNA = SQ.Mean)

Plate4_Telomeres<-print(subset_dfs[["telomeres"]])
```

```
## # A tibble: 72 x 23
##   X      Well Fluor Target      Content Sample Biological.Set.Name      Cq Cq.Mean
##   <lgl> <chr> <chr> <chr>      <chr>   <chr>   <lgl>          <dbl>  <dbl>
## 1 NA     A03     SYBR telomeres Unkn-01 7.1_B  NA      18.3    18.2
## 2 NA     A04     SYBR telomeres Unkn-01 7.1_B  NA      18.2    18.2
## 3 NA     A05     SYBR telomeres Unkn-09 15.5_B NA      18.4    18.4
## 4 NA     A06     SYBR telomeres Unkn-09 15.5_B NA      18.4    18.4
## 5 NA     A07     SYBR telomeres Unkn-17 17.2_D NA      18.1    18.1
## 6 NA     A08     SYBR telomeres Unkn-17 17.2_D NA      18.2    18.1
## 7 NA     A11     SYBR telomeres Unkn-33 15.3_C NA      17.9    18.1
## 8 NA     A12     SYBR telomeres Unkn-33 15.3_C NA      18.2    18.1
## 9 NA     B03     SYBR telomeres Unkn-02 3.2_C  NA      18.5    18.4
## 10 NA    B04     SYBR telomeres Unkn-02 3.2_C  NA      18.4    18.4
## # i 62 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lgl>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate4_Telomeres<-Plate4_Telomeres[,c("PlateID", "Well", "Sample", "Target", "Cq",
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate4_Telomeres <- Plate4_Telomeres %>%
  rename(Target_Telomeres = Target, Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean,
  Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)
  #rename(Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean, Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)

##### Make a final MPX dataset for by merging the Target datasets horizontally, in rows.
Plate4_FinalMPX <- merge(Plate4_SCNAG, Plate4_mtDNA, by = c("PlateID", "Well", "Sample"))

##### Normalize mtDNA
# Add a column called mtDNA, and calculate the normalized value
Plate4_FinalMPX$mtDNA <- (Plate4_FinalMPX$SQ.Mean_mtDNA / Plate4_FinalMPX$SQ.Mean_SCNAG)
```

```

# Recalculate mean across the replicates
Plate4_FinalMPX <- Plate4_FinalMPX %>%
  group_by(Sample) %>%
  mutate(
    mtDNA.Mean = mean(mtDNA)) %>%
  ungroup()

##### Merge final MPX with Telomeres
## Reduce datasets to single row per individual containing only the columns we want.
Plate4_FinalMPX <- distinct(Plate4_FinalMPX, PlateID, Sample, SQ.Mean_SCNAG,
Cq.Mean_SCNAG, SQ.Mean_mtDNA, mtDNA.Mean)
Plate4_FinalTelo <- distinct(Plate4_Telomeres, PlateID, Sample, SQ.Mean_Telomeres,
Cq.Mean_Telomeres)

## Merge the files horizontally
Plate4_FinalData <- merge(Plate4_FinalMPX, Plate4_FinalTelo, by = c("PlateID", "Sample"))

# Normalize Telomeres
Plate4_FinalData <- Plate4_FinalData %>% mutate(Telomeres.per.cell = SQ.Mean_Telomeres /
SQ.Mean_SCNAG)

# Assuming Plate4_FinalData has been prepared as per your previous steps

# Aggregate to get one row per sample, taking mean of normalized mtDNA and telomeres
Plate4_FinalData <- Plate4_FinalData %>%
  group_by(Sample) %>%
  summarize(
    SQ.Mean_SCNAG = mean(SQ.Mean_SCNAG),
    SQ.Mean_mtDNA = mean(SQ.Mean_mtDNA),
    mtDNA.Mean = mean(mtDNA.Mean),
    Cq.Mean_SCNAG = mean(Cq.Mean_SCNAG),
    SQ.Mean_Telomeres = mean(SQ.Mean_Telomeres),
    Cq.Mean_Telomeres = mean(Cq.Mean_Telomeres),
    Telomeres.per.cell = mean(Telomeres.per.cell)
  ) %>%
  ungroup()

# Write the final data file for this plate
write.csv(file = "./Data/Plate4_FinalData.csv", Plate4_FinalData, row.names = FALSE)

```

Plate 5

```

# Clear memory
rm(list=ls(all = TRUE))

# Import .csv files for each run for a particular plate of samples
Plate5_MPX <- read.csv("./Data/GT_Plate5_Multiplex_12_17_2024_Quantification_Cq_Results.csv")
dim (Plate5_MPX)

```

```
## [1] 160 16
```

```

Plate5_Telo <- read.csv("../Data/GT_Plate5_Telomeres_12_17_2024_Quantification_Cq_Results.csv")
dim (Plate5_Telo)

## [1] 76 16

# Concatenate data across runs for the same plate of samples
Plate5<- rbind(Plate5_MPX, Plate5_Telo)
dim(Plate5)

## [1] 236 16

# Add a column called PlateID and fill in correct Plate number to use as a variable in the statistics
Plate5$PlateID <- "Plate5"

# CHECK AND EDIT FOR YOUR DATA.
#Name Correct Targets based on the fluorophores used in your reaction.
Plate5$Target[Plate5$Fluor == "VIC"] <- "scnag"
Plate5$Target[Plate5$Fluor == "FAM"] <- "mtdna"
Plate5$Target[Plate5$Fluor == "SYBR"] <- "telomeres"

##### Look for outliers - 1st round #####

# Calculate the absolute difference from the mean Cq
Plate5$Diff_AVG_Cq <- abs(Plate5$Cq - Plate5$Cq.Mean)

# Identify outliers for mtdna and scnag samples based on >0.4 threshold
Plate5$Flag_outlier <- ifelse(Plate5$Target %in% c("mtdna", "scnag") &
Plate5$Diff_AVG_Cq > 0.4, "yes", "no")

# Identify outliers for telomeres based on >0.4 threshold
Plate5$Flag_outlier <- ifelse(Plate5$Target == "telomeres" & Plate5$Diff_AVG_Cq >
0.4, "yes", Plate5$Flag_outlier)

# Report and examine high Cq samples in the first round
HighCq <- Plate5[Plate5$Flag_outlier == "yes", c("Well", "Sample", "Fluor", "Diff_AVG_Cq")]
print(paste("Number of rows with high Cq in the first round:", nrow(HighCq)))

## [1] "Number of rows with high Cq in the first round: 4"

print(HighCq)

##      Well Sample Fluor Diff_AVG_Cq
## NA    <NA>    <NA>    <NA>      NA
## NA.1 <NA>    <NA>    <NA>      NA
## NA.2 <NA>    <NA>    <NA>      NA
## NA.3 <NA>    <NA>    <NA>      NA

# Remove outliers identified in the first round
Plate5 <- Plate5[Plate5$Flag_outlier != "yes", ]

# Verify the dimensions after removing outliers in the first round
print(dim(Plate5))

```

```
## [1] 236 19
```

```
##### Look for outliers - 2nd round #####

# Recalculate Cq Mean and Starting Quantity Mean based on remaining data
Plate5 <- Plate5 %>%
  group_by(Sample, Target) %>%
  mutate(
    Cq.Mean2 = mean(Cq),
    Sq.Mean2 = mean(Starting.Quantity..SQ.)
  ) %>%
  ungroup()

# Calculate the new difference from the updated mean Cq
Plate5$Diff_AVG_Cq_2 <- abs(Plate5$Cq - Plate5$Cq.Mean2)

# Identify outliers in the second round based on >0.4 threshold for telomeres
Plate5$Flag_outlier_2 <- ifelse(Plate5$Target == "telomeres" & Plate5$Diff_AVG_Cq_2 >
0.4, "yes", "no")

# Report and examine high Cq samples in the second round for telomeres
HighCq_2 <- Plate5[Plate5$Target == "telomeres" & Plate5$Diff_AVG_Cq_2 > 0.4,
c("Well", "Sample", "Fluor", "Diff_AVG_Cq_2")]
print(paste("Number of rows with high Cq in the second round for telomeres:", nrow(HighCq_2)))
```

```
## [1] "Number of rows with high Cq in the second round for telomeres: 4"
```

```
print(HighCq_2)
```

```
## # A tibble: 4 x 4
##   Well Sample Fluor Diff_AVG_Cq_2
##   <chr> <chr> <chr>      <dbl>
## 1 <NA> <NA> <NA>      NA
## 2 <NA> <NA> <NA>      NA
## 3 <NA> <NA> <NA>      NA
## 4 <NA> <NA> <NA>      NA
```

```
# Remove outliers identified in the second round for telomeres
Plate5 <- Plate5[!(Plate5$Target == "telomeres" & Plate5$Flag_outlier_2 == "yes"), ]

# Verify the dimensions after removing outliers in the second round
print(dim(Plate5))
```

```
## [1] 236 23
```

```
### Remove samples that do not have at least two rows
Plate5 <- Plate5 %>%
  group_by(Sample) %>%
  filter(n() >= 2) %>%
  ungroup()

# Final dimensions after all filtering steps
print(dim(Plate5))
```



```
## [1] 236 23
```

```
##### Remove negative controls and standards
# rows that have "NEG", "POS" in column "Sample" and remove rows with "STD" in Sample "Content"
Plate5 <- Plate5 %>%
  filter(!str_detect(Sample, "STD")) %>%
  filter(!str_detect(Sample, "NEG"))
dim(Plate5)
```

```
## [1] 198 23
```

```
##### Subset dataset based on the value in "Target" column
unique_targets <- unique(Plate5$Target)

# Create a list to store the subset dataframes
subset_dfs <- list()

# Loop through each unique value in 'Target', subset the dataframe, and store in subset_dfs
for (target_value in unique_targets) {
  subset_df <- subset(Plate5, Target == target_value)
  subset_dfs[[target_value]] <- subset_df
}

##### Now subset_dfs is a list where each element is a dataframe containing rows for each unique
Plate5_SCNAG<-print(subset_dfs[["scnag"]])
```

```
## # A tibble: 66 x 23
##   X      Well Fluor Target Content Sample Biological.Set.Name    Cq Cq.Mean
##   <lg1> <chr> <chr> <chr> <chr> <chr> <lg1>          <dbl> <dbl>
## 1 NA     A03   VIC   scnag Unkn-01 12.4_B NA          27.6  27.6
## 2 NA     A04   VIC   scnag Unkn-01 12.4_B NA          27.7  27.6
## 3 NA     A05   VIC   scnag Unkn-09 5.1_B  NA          28.1  28.1
## 4 NA     A06   VIC   scnag Unkn-09 5.1_B  NA          28.1  28.1
## 5 NA     A07   VIC   scnag Unkn-17 7.4_C  NA          27.1  27.2
## 6 NA     A08   VIC   scnag Unkn-17 7.4_C  NA          27.3  27.2
## 7 NA     A09   VIC   scnag Unkn-25 9.1_B  NA          27.5  27.4
## 8 NA     A10   VIC   scnag Unkn-25 9.1_B  NA          27.4  27.4
## 9 NA     A11   VIC   scnag Unkn-33 11.4_A NA          28.4  28.8
## 10 NA    A12   VIC   scnag Unkn-33 11.4_A NA          29.1  28.8
## # i 56 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate5_SCNAG<-Plate5_SCNAG[ ,c("PlateID", "Well", "Sample", "Target",
"Cq", "Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate5_SCNAG <- Plate5_SCNAG %>%
  rename(Target_SCNAG = Target, Cq_SCNAG = Cq, Cq.Mean_SCNAG = Cq.Mean,
  Flag_outlier_SCNAG=Flag_outlier, SQ.Mean_SCNAG=SQ.Mean)

Plate5_mtDNA<-print(subset_dfs[["mtdna"]])
```

```
## # A tibble: 66 x 23
##   X      Well Fluor Target Content Sample Biological.Set.Name    Cq Cq.Mean
##   <lg1> <chr> <chr> <chr>   <chr>   <chr>   <lg1>          <dbl>   <dbl>
## 1 NA     A03   FAM   mtdna  Unkn-01 12.4_B NA          22.2    22.2
## 2 NA     A04   FAM   mtdna  Unkn-01 12.4_B NA          22.2    22.2
## 3 NA     A05   FAM   mtdna  Unkn-09 5.1_B NA          21.8    21.8
## 4 NA     A06   FAM   mtdna  Unkn-09 5.1_B NA          21.9    21.8
## 5 NA     A07   FAM   mtdna  Unkn-17 7.4_C NA          21.6    21.7
## 6 NA     A08   FAM   mtdna  Unkn-17 7.4_C NA          21.8    21.7
## 7 NA     A09   FAM   mtdna  Unkn-25 9.1_B NA          21.4    21.4
## 8 NA     A10   FAM   mtdna  Unkn-25 9.1_B NA          21.4    21.4
## 9 NA     A11   FAM   mtdna  Unkn-33 11.4_A NA         23.1    23.2
## 10 NA    A12   FAM   mtdna  Unkn-33 11.4_A NA         23.2    23.2
## # i 56 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate5_mtDNA<-Plate5_mtDNA[ ,c("PlateID", "Well", "Sample", "Target", "Cq",
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
```

```
Plate5_mtDNA <- Plate5_mtDNA %>%
  rename(Target_mtDNA = Target, Cq_mtDNA = Cq, Cq.Mean_mtDNA = Cq.Mean,
  Flag_outlier_mtDNA = Flag_outlier, SQ.Mean_mtDNA = SQ.Mean)
```

```
Plate5_Telomeres<-print(subset_dfs[["telomeres"]])
```

```
## # A tibble: 66 x 23
##   X      Well Fluor Target    Content Sample Biological.Set.Name    Cq Cq.Mean
##   <lg1> <chr> <chr> <chr>      <chr>   <chr>   <lg1>          <dbl>   <dbl>
## 1 NA     A03   SYBR telomeres Unkn-01 12.4_B NA          19.2    19.2
## 2 NA     A04   SYBR telomeres Unkn-01 12.4_B NA          19.1    19.2
## 3 NA     A05   SYBR telomeres Unkn-09 5.1_B NA          18.9    18.9
## 4 NA     A06   SYBR telomeres Unkn-09 5.1_B NA          18.9    18.9
## 5 NA     A07   SYBR telomeres Unkn-17 7.4_C NA          18.8    18.8
## 6 NA     A08   SYBR telomeres Unkn-17 7.4_C NA          18.8    18.8
## 7 NA     A09   SYBR telomeres Unkn-25 9.1_B NA          18.7    18.7
## 8 NA     A10   SYBR telomeres Unkn-25 9.1_B NA          18.7    18.7
## 9 NA     B03   SYBR telomeres Unkn-02 11.1_D NA          19.1    19.2
## 10 NA    B04   SYBR telomeres Unkn-02 11.1_D NA          19.3    19.2
## # i 56 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate5_Telomeres<-Plate5_Telomeres[ ,c("PlateID", "Well", "Sample", "Target",
"Cq", "Cq.Mean", "Flag_outlier", "SQ.Mean")]
```

```
Plate5_Telomeres <- Plate5_Telomeres %>%
  rename(Target_Telomeres = Target, Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean,
  Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)
```

```

#rename(Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean, Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)

#####. Make a final MPX dataset for by merging the Target datasets horizontally, in rows.
Plate5_FinalMPX <- merge(Plate5_SCNAG, Plate5_mtDNA, by = c("PlateID", "Well", "Sample"))

##### Normalize mtDNA
# Add a column called mtDNA, and calculate the normalized value
Plate5_FinalMPX$mtDNA <- (Plate5_FinalMPX$SQ.Mean_mtDNA / Plate5_FinalMPX$SQ.Mean_SCNAG)

# Recalculate mean across the replicates
Plate5_FinalMPX <- Plate5_FinalMPX %>%
  group_by(Sample) %>%
  mutate(
    mtDNA.Mean = mean(mtDNA)) %>%
  ungroup()

##### Merge final MPX with Telomeres
## Reduce datasets to single row per individual containing only the columns we want.
Plate5_FinalMPX <- distinct(Plate5_FinalMPX, PlateID, Sample, SQ.Mean_SCNAG,
  Cq.Mean_SCNAG, SQ.Mean_mtDNA, mtDNA.Mean)
Plate5_FinalTelo <- distinct(Plate5_Telomeres, PlateID, Sample, SQ.Mean_Telomeres,
  Cq.Mean_Telomeres)

## Merge the files horizontally
Plate5_FinalData <- merge(Plate5_FinalMPX, Plate5_FinalTelo, by = c("PlateID", "Sample"))

# Normalize Telomeres
Plate5_FinalData <- Plate5_FinalData %>% mutate(Telomeres.per.cell = SQ.Mean_Telomeres /
  SQ.Mean_SCNAG)

# Assuming Plate5_FinalData has been prepared as per your previous steps

# Aggregate to get one row per sample, taking mean of normalized mtDNA and telomeres
Plate5_FinalData <- Plate5_FinalData %>%
  group_by(Sample) %>%
  summarize(
    SQ.Mean_SCNAG = mean(SQ.Mean_SCNAG),
    SQ.Mean_mtDNA = mean(SQ.Mean_mtDNA),
    mtDNA.Mean = mean(mtDNA.Mean),
    Cq.Mean_SCNAG = mean(Cq.Mean_SCNAG),
    SQ.Mean_Telomeres = mean(SQ.Mean_Telomeres),
    Cq.Mean_Telomeres = mean(Cq.Mean_Telomeres),
    Telomeres.per.cell = mean(Telomeres.per.cell)
  ) %>%
  ungroup()

# Write the final data file for this plate
write.csv(file = "./Data/Plate5_FinalData.csv", Plate5_FinalData, row.names = FALSE)

```

Plate 6

```
# Clear memory
rm(list=ls(all = TRUE))

# Import .csv files for each run for a particular plate of samples
Plate6_MPX <- read.csv("./Data/GT_Plate6_Multiplex_12_18_2024_Quantification_Cq_Results.csv")
dim (Plate6_MPX)

## [1] 148 16

Plate6_MPX <- Plate6_MPX[, -c(1)]

Plate6_Telo <- read.csv("./Data/GT_Plate6_Telomeres_12_18_2024_Quantification_Cq_Results.csv")
dim (Plate6_Telo)

## [1] 72 15

# Concatenate data across runs for the same plate of samples
Plate6<- rbind(Plate6_MPX, Plate6_Telo)
dim(Plate6)

## [1] 220 15

# Add a column called PlateID and fill in correct Plate number to use as a variable in the statistics
Plate6$PlateID <- "Plate6"

# CHECK AND EDIT FOR YOUR DATA.
#Name Correct Targets based on the fluorophores used in your reaction.
Plate6$Target[Plate6$Fluor == "VIC"] <- "scnag"
Plate6$Target[Plate6$Fluor == "FAM"] <- "mtdna"
Plate6$Target[Plate6$Fluor == "SYBR"] <- "telomeres"

##### Look for outliers - 1st round #####

# Calculate the absolute difference from the mean Cq
Plate6$Diff_AVG_Cq <- abs(Plate6$Cq - Plate6$Cq.Mean)

# Identify outliers for mtdna and scnag samples based on >0.4 threshold
Plate6$Flag_outlier <- ifelse(Plate6$Target %in% c("mtdna", "scnag") &
Plate6$Diff_AVG_Cq > 0.4, "yes", "no")

# Identify outliers for telomeres based on >0.4 threshold
Plate6$Flag_outlier <- ifelse(Plate6$Target == "telomeres" & Plate6$Diff_AVG_Cq >
0.4, "yes", Plate6$Flag_outlier)

# Report and examine high Cq samples in the first round
HighCq <- Plate6[Plate6$Flag_outlier == "yes", c("Well", "Sample", "Fluor", "Diff_AVG_Cq")]
print(paste("Number of rows with high Cq in the first round:", nrow(HighCq)))

## [1] "Number of rows with high Cq in the first round: 11"
```

```
print(HighCq)
```

```
##      Well Sample Fluor Diff_AVG_Cq
## NA    <NA>    <NA>    <NA>        NA
## NA.1 <NA>    <NA>    <NA>        NA
## 123   F01   STD6   VIC    0.4147679
## 124   F02   STD6   VIC    0.4147679
## NA.2 <NA>    <NA>    <NA>        NA
## 191   E11   9.2_D  SYBR    0.5102138
## 192   E12   9.2_D  SYBR    0.5102138
## 195   F05   5.4_A  SYBR    1.2430682
## 196   F06   5.4_A  SYBR    1.2430682
## 201   F11  13.2_A  SYBR    0.5140238
## 202   F12  13.2_A  SYBR    0.5140238
```

```
# Remove outliers identified in the first round
Plate6 <- Plate6[Plate6$Flag_outlier != "yes", ]
```

```
# Verify the dimensions after removing outliers in the first round
print(dim(Plate6))
```

```
## [1] 212  18
```

```
##### Look for outliers - 2nd round #####
```

```
# Recalculate Cq Mean and Starting Quantity Mean based on remaining data
Plate6 <- Plate6 %>%
```

```
  group_by(Sample, Target) %>%
  mutate(
    Cq.Mean2 = mean(Cq),
    Sq.Mean2 = mean(Starting.Quantity..SQ.)
  ) %>%
  ungroup()
```

```
# Calculate the new difference from the updated mean Cq
Plate6$Diff_AVG_Cq_2 <- abs(Plate6$Cq - Plate6$Cq.Mean2)
```

```
# Identify outliers in the second round based on >0.4 threshold for telomeres
Plate6$Flag_outlier_2 <- ifelse(Plate6$Target == "telomeres" & Plate6$Diff_AVG_Cq_2 >
0.4, "yes", "no")
```

```
# Report and examine high Cq samples in the second round for telomeres
HighCq_2 <- Plate6[Plate6$Target == "telomeres" & Plate6$Diff_AVG_Cq_2 > 0.4,
c("Well", "Sample", "Fluor", "Diff_AVG_Cq_2")]
print(paste("Number of rows with high Cq in the second round for telomeres:", nrow(HighCq_2)))
```

```
## [1] "Number of rows with high Cq in the second round for telomeres: 3"
```

```
print(HighCq_2)
```

```
## # A tibble: 3 x 4
```

```
## Well Sample Fluor Diff_AVG_Cq_2
## <chr> <chr> <chr> <dbl>
## 1 <NA> <NA> <NA> NA
## 2 <NA> <NA> <NA> NA
## 3 <NA> <NA> <NA> NA
```

```
# Remove outliers identified in the second round for telomeres
Plate6 <- Plate6[!(Plate6$Target == "telomeres" & Plate6$Flag_outlier_2 == "yes"), ]

# Verify the dimensions after removing outliers in the second round
print(dim(Plate6))
```

```
## [1] 212 22
```

```
### Remove samples that do not have at least two rows
Plate6 <- Plate6 %>%
  group_by(Sample) %>%
  filter(n() >= 2) %>%
  ungroup()
```

```
# Final dimensions after all filtering steps
print(dim(Plate6))
```

```
## [1] 212 22
```

```
##### Remove negative controls and standards
# rows that have "NEG", "POS" in column "Sample" and remove rows with "STD" in Sample "Content"
Plate6 <- Plate6 %>%
  filter(!str_detect(Sample, "STD")) %>%
  filter(!str_detect(Sample, "NEG"))
dim(Plate6)
```

```
## [1] 176 22
```

```
##### Subset dataset based on the value in "Target" column
unique_targets <- unique(Plate6$Target)
```

```
# Create a list to store the subset dataframes
subset_dfs <- list()
```

```
# Loop through each unique value in 'Target', subset the dataframe, and store in subset_dfs
for (target_value in unique_targets) {
  subset_df <- subset(Plate6, Target == target_value)
  subset_dfs[[target_value]] <- subset_df
}
```

```
##### Now subset_dfs is a list where each element is a dataframe containing rows for each unique Target
Plate6_SCNAG<-print(subset_dfs[["scnag"]])
```

```
## # A tibble: 60 x 22
```

```
## Well Fluor Target Content Sample Biological.Set.Name Cq Cq.Mean
```

```
##      <chr> <chr> <chr>      <chr>      <chr>      <chr>      <dbl>      <dbl>
## 1 A07    VIC    scnag    Unkn-17  9.2_B  NA                27.2      27.2
## 2 A08    VIC    scnag    Unkn-17  9.2_B  NA                27.3      27.2
## 3 A09    VIC    scnag    Unkn-25 10.3_C NA                27.5      27.6
## 4 A10    VIC    scnag    Unkn-25 10.3_C NA                27.6      27.6
## 5 A11    VIC    scnag    Unkn-33 11.4_B NA                27.3      27.3
## 6 A12    VIC    scnag    Unkn-33 11.4_B NA                27.4      27.3
## 7 B03    VIC    scnag    Unkn-02 15.1_A NA                25.4      25.6
## 8 B04    VIC    scnag    Unkn-02 15.1_A NA                25.8      25.6
## 9 B05    VIC    scnag    Unkn-10 10.5_C NA                26.7      26.8
## 10 B06   VIC    scnag    Unkn-10 10.5_C NA                26.8      26.8
## # i 50 more rows
## # i 14 more variables: Cq.Std..Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std..Dev <dbl>,
## #   Set.Point <int>, Well.Note <lgl>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate6_SCNAG<-Plate6_SCNAG[ ,c("PlateID", "Well", "Sample", "Target", "Cq",
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate6_SCNAG <- Plate6_SCNAG %>%
  rename(Target_SCNAG = Target, Cq_SCNAG = Cq, Cq.Mean_SCNAG = Cq.Mean,
  Flag_outlier_SCNAG=Flag_outlier, SQ.Mean_SCNAG=SQ.Mean)

Plate6_mtDNA<-print(subset_dfs[["mtdna"]])
```

```
## # A tibble: 60 x 22
##   Well Fluor Target Content Sample Biological.Set.Name      Cq Cq.Mean
##   <chr> <chr> <chr>      <chr>      <chr>      <lgl>      <dbl>      <dbl>
## 1 A07    FAM    mtdna    Unkn-17  9.2_B  NA                21.6      21.6
## 2 A08    FAM    mtdna    Unkn-17  9.2_B  NA                21.6      21.6
## 3 A09    FAM    mtdna    Unkn-25 10.3_C NA                22.4      22.5
## 4 A10    FAM    mtdna    Unkn-25 10.3_C NA                22.5      22.5
## 5 A11    FAM    mtdna    Unkn-33 11.4_B NA                22.0      22.0
## 6 A12    FAM    mtdna    Unkn-33 11.4_B NA                22.0      22.0
## 7 B03    FAM    mtdna    Unkn-02 15.1_A NA                20.0      20.2
## 8 B04    FAM    mtdna    Unkn-02 15.1_A NA                20.3      20.2
## 9 B05    FAM    mtdna    Unkn-10 10.5_C NA                21.5      21.5
## 10 B06   FAM    mtdna    Unkn-10 10.5_C NA                21.5      21.5
## # i 50 more rows
## # i 14 more variables: Cq.Std..Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std..Dev <dbl>,
## #   Set.Point <int>, Well.Note <lgl>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate6_mtDNA<-Plate6_mtDNA[ ,c("PlateID", "Well", "Sample", "Target", "Cq",
"Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate6_mtDNA <- Plate6_mtDNA %>%
  rename(Target_mtDNA = Target, Cq_mtDNA = Cq, Cq.Mean_mtDNA = Cq.Mean,
  Flag_outlier_mtDNA = Flag_outlier, SQ.Mean_mtDNA = SQ.Mean)

Plate6_Telomeres<-print(subset_dfs[["telomeres"]])
```

```
## # A tibble: 56 x 22
##   Well Fluor Target Content Sample Biological.Set.Name Cq Cq.Mean
##   <chr> <chr> <chr>   <chr>   <chr>   <lg1>         <dbl>   <dbl>
## 1 A07 SYBR telomeres Unkn-17 9.2_B NA          17.3    17.3
## 2 A08 SYBR telomeres Unkn-17 9.2_B NA          17.4    17.3
## 3 A09 SYBR telomeres Unkn-25 10.3_C NA         18.1    18.1
## 4 A10 SYBR telomeres Unkn-25 10.3_C NA         18.0    18.1
## 5 A11 SYBR telomeres Unkn-33 11.4_B NA          17.5    17.7
## 6 A12 SYBR telomeres Unkn-33 11.4_B NA          17.9    17.7
## 7 B05 SYBR telomeres Unkn-10 10.5_C NA          17.1    17.1
## 8 B06 SYBR telomeres Unkn-10 10.5_C NA          17.2    17.1
## 9 B07 SYBR telomeres Unkn-18 7.5_B NA          17.3    17.3
## 10 B08 SYBR telomeres Unkn-18 7.5_B NA          17.4    17.3
## # i 46 more rows
## # i 14 more variables: Cq.Std.Dev <dbl>, Starting.Quantity..SQ. <dbl>,
## #   Log.Starting.Quantity <dbl>, SQ.Mean <dbl>, SQ.Std.Dev <dbl>,
## #   Set.Point <int>, Well.Note <lg1>, PlateID <chr>, Diff_AVG_Cq <dbl>,
## #   Flag_outlier <chr>, Cq.Mean2 <dbl>, Sq.Mean2 <dbl>, Diff_AVG_Cq_2 <dbl>,
## #   Flag_outlier_2 <chr>
```

```
Plate6_Telomeres<-Plate6_Telomeres[,c("PlateID", "Well", "Sample", "Target",
"Cq", "Cq.Mean", "Flag_outlier", "SQ.Mean")]
Plate6_Telomeres <- Plate6_Telomeres %>%
  rename(Target_Telomeres = Target, Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean,
  Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)
#rename(Cq_Telomeres = Cq, Cq.Mean_Telomeres = Cq.Mean, Flag_outlier_Telomeres = Flag_outlier, SQ.Mean_Telomeres = SQ.Mean)

##### Make a final MPX dataset for by merging the Target datasets horizontally, in rows.
Plate6_FinalMPX <- merge(Plate6_SCNAG, Plate6_mtDNA, by = c("PlateID", "Well", "Sample"))

##### Normalize mtDNA
# Add a column called mtDNA, and calculate the normalized value
Plate6_FinalMPX$mtDNA <- (Plate6_FinalMPX$SQ.Mean_mtDNA / Plate6_FinalMPX$SQ.Mean_SCNAG)

# Recalculate mean across the replicates
Plate6_FinalMPX <- Plate6_FinalMPX %>%
  group_by(Sample) %>%
  mutate(
    mtDNA.Mean = mean(mtDNA)) %>%
  ungroup()

##### Merge final MPX with Telomeres
## Reduce datasets to single row per individual containing only the columns we want.
Plate6_FinalMPX <- distinct(Plate6_FinalMPX, PlateID, Sample, SQ.Mean_SCNAG,
Cq.Mean_SCNAG, SQ.Mean_mtDNA, mtDNA.Mean)
Plate6_FinalTelo <- distinct(Plate6_Telomeres, PlateID, Sample, SQ.Mean_Telomeres,
Cq.Mean_Telomeres)

## Merge the files horizontally
Plate6_FinalData <- merge(Plate6_FinalMPX, Plate6_FinalTelo, by = c("PlateID", "Sample"))

# Normalize Telomeres
Plate6_FinalData <- Plate6_FinalData %>% mutate(Telomeres.per.cell = SQ.Mean_Telomeres /
SQ.Mean_SCNAG)
```



```

# Assuming Plate6_FinalData has been prepared as per your previous steps

# Aggregate to get one row per sample, taking mean of normalized mtDNA and telomeres
Plate6_FinalData <- Plate6_FinalData %>%
  group_by(Sample) %>%
  summarize(
    SQ.Mean_SCNAG = mean(SQ.Mean_SCNAG),
    SQ.Mean_mtDNA = mean(SQ.Mean_mtDNA),
    mtDNA.Mean = mean(mtDNA.Mean),
    Cq.Mean_SCNAG = mean(Cq.Mean_SCNAG),
    SQ.Mean_Telomeres = mean(SQ.Mean_Telomeres),
    Cq.Mean_Telomeres = mean(Cq.Mean_Telomeres),
    Telomeres.per.cell = mean(Telomeres.per.cell)
  ) %>%
  ungroup()

# Write the final data file for this plate
write.csv(file = "./Data/Plate6_FinalData.csv", Plate6_FinalData, row.names = FALSE)

```

Now that we have all the qPCR plates processed and we have filtered samples based on Cq values, we will merge all the plates together so that we have values corresponding to normalized telomere length and normalized mtDNA density corresponding to an individual into one file.

```

Plate1_FinalData <- read.csv("./Data/Plate1_FinalData.csv")
Plate2_FinalData <- read.csv("./Data/Plate2_FinalData.csv")
Plate3_FinalData <- read.csv("./Data/Plate3_FinalData.csv")
Plate4_FinalData <- read.csv("./Data/Plate4_FinalData.csv")
Plate5_FinalData <- read.csv("./Data/Plate5_FinalData.csv")
Plate6_FinalData <- read.csv("./Data/Plate6_FinalData.csv")

# Concatenate the datasets
qPCR_FinalData <- rbind(Plate1_FinalData, Plate2_FinalData, Plate3_FinalData,
  Plate4_FinalData, Plate5_FinalData, Plate6_FinalData)

# Verify the dimensions of the combined data
print(dim(qPCR_FinalData))

```

```
## [1] 205    8
```

Then, we will merge the qPCR data with the Trait data, where we have information on Plate_ID, Treatment and Timepoint.

```
Trait <- read.csv("../Data/Trait_MetaData.csv")
dim (Trait)
```

```
## [1] 240 4
```

```
dim(qPCR_FinalData)
```

```
## [1] 205 8
```

```
# Merge both datasets
FinalData <- merge(qPCR_FinalData, Trait, by = c("Sample"))

# Save the merged data to a final CSV file
write.csv(FinalData, "../Data/GT_FinalData.csv", row.names = FALSE)

# Optional: Print the dimensions of the final merged data
print(dim(FinalData))
```

```
## [1] 205 11
```

```
##### Calculating the percent change in mtDNA and Telomere length #####
```

```
##### Prepare the data #####
```

```
# Load the data
data <- read_csv("../Data/GT_FinalData.csv")

# Creating a new column called Tortoise_ID by extracting the individual ID from the column Sample;
# sub() is a base R function that replaces the first match of a pattern in a string.
# "_.*" is a regular expression that means: "an underscore (_) followed by anything (.*)"
# "" means: replace that part with nothing (i.e., remove it).
```

```
data <- data %>%
  mutate(Tortoise_ID = sub("_.*", "", Sample))
```

```
# Arrange by Tortoise_ID and Time_Point (A to D)
data <- data %>%
  arrange(Tortoise_ID, Time_Point)
```

```
# Filter data for Interval A to B
interval_A_to_B <- data %>%
  filter(Time_Point %in% c("A", "B"))
```

```
# Filter data for Interval B to C
interval_B_to_C <- data %>%
  filter(Time_Point %in% c("B", "C"))
```

```
# Filter data for Interval C to D
interval_C_to_D <- data %>%
  filter(Time_Point %in% c("C", "D"))
```

Calculate the percent change in telomere length and mtDNA from one timepoint to the next. Timepoints are: A <-> B (Winter), B <-> C (Early Spring) and C <-> D (Late Spring).

```
interval_A_to_B <- interval_A_to_B %>%
  group_by(Tortoise_ID) %>%
  mutate(
    mtDNA_percent_change = (mtDNA.Mean - lag(mtDNA.Mean)) / lag(mtDNA.Mean) * 100,
    Telomeres_percent_change = (Telomeres.per.cell - lag(Telomeres.per.cell)) /
    lag(Telomeres.per.cell) * 100,
    Transition = paste0(lag(Time_Point), "_to_", Time_Point)
  ) %>%
  filter(!is.na(mtDNA_percent_change), !is.na(Telomeres_percent_change)) %>%
  ungroup()

# Calculate percent changes from one timepoint to the next, i.e. from timepoint B to C
interval_B_to_C <- interval_B_to_C %>%
  group_by(Tortoise_ID) %>%
  mutate(
    mtDNA_percent_change = (mtDNA.Mean - lag(mtDNA.Mean)) / lag(mtDNA.Mean) * 100,
    Telomeres_percent_change = (Telomeres.per.cell - lag(Telomeres.per.cell)) /
    lag(Telomeres.per.cell) * 100,
    Transition = paste0(lag(Time_Point), "_to_", Time_Point)
  ) %>%
  filter(!is.na(mtDNA_percent_change), !is.na(Telomeres_percent_change)) %>%
  ungroup()

# Calculate percent changes from one timepoint to the next, i.e. from timepoint C to D
interval_C_to_D <- interval_C_to_D %>%
  group_by(Tortoise_ID) %>%
  mutate(
    mtDNA_percent_change = (mtDNA.Mean - lag(mtDNA.Mean)) / lag(mtDNA.Mean) * 100,
    Telomeres_percent_change = (Telomeres.per.cell - lag(Telomeres.per.cell)) /
    lag(Telomeres.per.cell) * 100,
    Transition = paste0(lag(Time_Point), "_to_", Time_Point)
  ) %>%
  filter(!is.na(mtDNA_percent_change), !is.na(Telomeres_percent_change)) %>%
  ungroup()

# Merge all intervals together
merged_intervals <- rbind(interval_A_to_B, interval_B_to_C, interval_C_to_D)

# Now pivot to wide format
percent_change_wide <- merged_intervals %>%
  select(Tortoise_ID, Transition, mtDNA_percent_change, Telomeres_percent_change) %>% # selecting those
  pivot_wider(
    names_from = Transition,
    values_from = c(mtDNA_percent_change, Telomeres_percent_change),
    names_glue = "{.value}_{Transition}" #.value refers to the column names from the values_from argument
    #Transition refers to the unique values in the Transition column, which will become part of the new
  )
```

```
# Produce a .csv file with one row per Tortoise_ID and each column represents a change in either mtDNA
write_csv(percent_change_wide, "./Data/GT_PercentChange.csv")
```

Finally, we will join the Percent Change file with the Growth Rate file to run the mixed-effect model.

```
growth_data <- read_csv("./Data/GT_GrowthData.csv") %>%
  filter(!(Tortoise_ID %in% c("Not_Viable", "GT2023_N05.03", "GT2023_N06.01",
    "GT2023_N05.06", "GT2023_N15.04"))) %>%
  drop_na() # removes any remaining rows with NA

# Split the ID into two parts, pad both to two digits, and paste back together
percent_change_wide <- read_csv("./Data/GT_PercentChange.csv") %>%
  mutate(
    Tortoise_ID = sprintf("GT2023_N%02d.%02d",
      as.integer(sub("\\..*", "", Tortoise_ID)), # part before the decimal
      as.integer(sub(".*\\. ", "", Tortoise_ID)) # part after the decimal
    )
  )

# Now merge the two datasets by Tortoise_ID
merged_data <- left_join(growth_data, percent_change_wide, by = "Tortoise_ID")

# Produce a merged .csv file containing growth rate and percent change in mtDNA and telomere length from
write_csv(merged_data, "./Data/GT_Growth_Telo_mtDNA.csv")
```

Data Analysis (Telomere length)

We will run three mixed-effect models at each interval with Percent change in Telomere length as a response variable and growth rate and treatment as fixed effects, and Nest_ID as a random effect.

Hypothesis: Hatchling tortoises experiencing early life fast growth at constant warm temperature will have shorter telomere length in blood cells 3 months post-dormancy compared to animals that experienced cold dormancy?

```
merged_data <- read_csv("./Data/GT_Growth_Telo_mtDNA.csv", na.strings = "NA")
str(merged_data)
```

```
## 'data.frame':   56 obs. of  13 variables:
## $ Nest_ID      : chr  "GT2023_N02" "GT2023_N02" "GT2023_N02" "GT2023_N03" ...
## $ Treatment    : chr  "Cold-Dormancy" "Cold-Dormancy" "Constant-Warmth" "Cold-Dormancy" ...
## $ Tortoise_ID  : chr  "GT2023_N02.02" "GT2023_N02.03" "GT2023_N02.04" "GT2023_N03.01" ...
## $ Tank         : int   3 11 5 6 1 8 9 10 11 1 ...
## $ Growth_rate_During : num  0.0873 0.0232 0.7136 0.0687 0.1145 ...
## $ Growth_rate_3_Weeks_Post : num  0.547 0.626 1.037 0.789 0.479 ...
```

```
## $ Growth_rate_3_Months_Post      : num  0.784 0.46 0.757 0.929 0.774 ...
## $ mtDNA_percent_change_A_to_B    : num  NA 22.7 52.4 121.4 25.5 ...
## $ mtDNA_percent_change_B_to_C    : num  -0.3 19.9 -2.22 -15.88 -31.53 ...
## $ mtDNA_percent_change_C_to_D    : num  10.41 -47.21 -37.05 -43.76 -7.06 ...
## $ Telomeres_percent_change_A_to_B: num  NA 36.6 254.4 39.9 379 ...
## $ Telomeres_percent_change_B_to_C: num  -6.9 99.4 -48.6 64.6 -76.5 ...
## $ Telomeres_percent_change_C_to_D: num  117.07 -49.67 -43.96 -16.96 5.37 ...

##### Correlation for telomeres Winter (between Before and During) #####
# Subset to remove NAs for interval A to B
subset_telo_ab <- subset(merged_data, !is.na(Telomeres_percent_change_A_to_B))

# Overall correlation
cor.test(subset_telo_ab$Telomeres_percent_change_A_to_B, subset_telo_ab$Growth_rate_During,
method = "spearman")

##
## Spearman's rank correlation rho
##
## data: subset_telo_ab$Telomeres_percent_change_A_to_B and subset_telo_ab$Growth_rate_During
## S = 9426, p-value = 0.4755
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.1157598

# Correlation by treatment
by(subset_telo_ab, subset_telo_ab$Treatment, function(df) {
  cor.test(df$Telomeres_percent_change_A_to_B, df$Growth_rate_During, method = "spearman")
})

## subset_telo_ab$Treatment: Cold-Dormancy
##
## Spearman's rank correlation rho
##
## data: df$Telomeres_percent_change_A_to_B and df$Growth_rate_During
## S = 1386, p-value = 0.6655
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.1
##
## -----
## subset_telo_ab$Treatment: Constant-Warmth
##
## Spearman's rank correlation rho
##
## data: df$Telomeres_percent_change_A_to_B and df$Growth_rate_During
## S = 814, p-value = 0.2345
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.2859649
```

```
##### Correlation for telomeres Early Spring (between During and 3 Weeks Post)
# Subset to remove NAs for interval B to C
subset_telo_bc <- subset(merged_data, !is.na(Telomeres_percent_change_B_to_C))

# Overall correlation
cor.test(subset_telo_bc$Telomeres_percent_change_B_to_C, subset_telo_bc$Growth_rate_3_Weeks_Post,
method = "spearman")

##
## Spearman's rank correlation rho
##
## data: subset_telo_bc$Telomeres_percent_change_B_to_C and subset_telo_bc$Growth_rate_3_Weeks_Post
## S = 27019, p-value = 0.5248
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## -0.08930731

# Correlation by treatment
by(subset_telo_bc, subset_telo_bc$Treatment, function(df) {
  cor.test(df$Telomeres_percent_change_B_to_C, df$Growth_rate_3_Weeks_Post, method = "spearman")
})

## subset_telo_bc$Treatment: Cold-Dormancy
##
## Spearman's rank correlation rho
##
## data: df$Telomeres_percent_change_B_to_C and df$Growth_rate_3_Weeks_Post
## S = 3281, p-value = 0.994
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## -0.001526718
##
## -----
## subset_telo_bc$Treatment: Constant-Warmth
##
## Spearman's rank correlation rho
##
## data: df$Telomeres_percent_change_B_to_C and df$Growth_rate_3_Weeks_Post
## S = 2995, p-value = 0.9076
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## -0.02393572

##### Correlation for telomeres Late Spring (Between 3 Weeks Post and 3 Months Post)
# Subset to remove NAs for interval C to D
subset_telo_cd <- subset(merged_data, !is.na(Telomeres_percent_change_C_to_D))

# Overall correlation
cor.test(subset_telo_cd$Telomeres_percent_change_C_to_D, subset_telo_cd$Growth_rate_3_Months_Post,
method = "spearman")
```

```
##
## Spearman's rank correlation rho
##
## data: subset_telo_cd$Telomeres_percent_change_C_to_D and subset_telo_cd$Growth_rate_3_Months_Post
## S = 23352, p-value = 0.9823
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.003158951
```

```
# Correlation by treatment
by(subset_telo_cd, subset_telo_cd$Treatment, function(df) {
  cor.test(df$Telomeres_percent_change_C_to_D, df$Growth_rate_3_Months_Post,
    method = "spearman")
})
```

```
## subset_telo_cd$Treatment: Cold-Dormancy
##
## Spearman's rank correlation rho
##
## data: df$Telomeres_percent_change_C_to_D and df$Growth_rate_3_Months_Post
## S = 3276, p-value = 1
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0
##
## -----
## subset_telo_cd$Treatment: Constant-Warmth
##
## Spearman's rank correlation rho
##
## data: df$Telomeres_percent_change_C_to_D and df$Growth_rate_3_Months_Post
## S = 2612, p-value = 0.9839
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## -0.004615385
```

```
# Running a linear-regression for A to B interval
model_telo_ab <- lmer(Telomeres_percent_change_A_to_B ~ Growth_rate_During + Treatment +
  (1|Nest_ID), data = subset_telo_ab)
summary(model_telo_ab)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Telomeres_percent_change_A_to_B ~ Growth_rate_During + Treatment +
##      (1 | Nest_ID)
##      Data: subset_telo_ab
##
## REML criterion at convergence: 460.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -0.9824 -0.5294 -0.3318 -0.0025 3.6252
##
## Random effects:
## Groups Name Variance Std.Dev.
## Nest_ID (Intercept) 0 0.0
## Residual 13028 114.1
## Number of obs: 40, groups: Nest_ID, 14
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 38.34 27.81 1.379
## Growth_rate_During 74.25 160.30 0.463
## TreatmentConstant-Warmth -39.15 97.85 -0.400
##
## Correlation of Fixed Effects:
## (Intr) Grw__D
## Grwth_rt_Dr -0.445
## TrtmntCns-W 0.185 -0.929
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
emmeans(model_telo_ab, pairwise ~ Treatment)
```

```
## $emmeans
## Treatment emmean SE df lower.CL upper.CL
## Cold-Dormancy 64.1 51.9 34.0 -41.3 169
## Constant-Warmth 24.9 57.8 30.9 -93.0 143
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## (Cold-Dormancy) - (Constant-Warmth) 39.2 103 35.6 0.379 0.7067
##
## Degrees-of-freedom method: kenward-roger
```

```
# Running a linear-regression for B to C interval
model_telo_bc <- lmer(Telomeres_percent_change_B_to_C ~ Growth_rate_3_Weeks_Post +
Treatment + (1|Nest_ID), data = subset_telo_bc)
summary(model_telo_bc)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Telomeres_percent_change_B_to_C ~ Growth_rate_3_Weeks_Post +
## Treatment + (1 | Nest_ID)
## Data: subset_telo_bc
##
## REML criterion at convergence: 579
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -1.4587 -0.6634 -0.2165 0.5852 2.6614
##
```



```

## Random effects:
## Groups Name Variance Std.Dev.
## Nest_ID (Intercept) 0 0.00
## Residual 5279 72.66
## Number of obs: 53, groups: Nest_ID, 14
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 24.897 22.123 1.125
## Growth_rate_3_Weeks_Post 3.738 26.829 0.139
## TreatmentConstant-Warmth -19.025 25.522 -0.745
##
## Correlation of Fixed Effects:
## (Intr) G__3_W
## Grwt__3_W_P -0.775
## TrtmntCns-W 0.136 -0.623
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

emmeans(model_telo_bc, pairwise ~ Treatment)

## $emmeans
## Treatment emmean SE df lower.CL upper.CL
## Cold-Dormancy 28.37 16.3 39.3 -4.52 61.3
## Constant-Warmth 9.35 16.9 33.0 -25.05 43.7
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## (Cold-Dormancy) - (Constant-Warmth) 19 26.3 48.8 0.724 0.4726
##
## Degrees-of-freedom method: kenward-roger

# Running a linear-regression for C to D interval
model_telo_cd <- lmer(Telomeres_percent_change_C_to_D ~ Growth_rate_3_Months_Post +
Treatment + (1|Nest_ID), data = subset_telo_cd)
summary(model_telo_cd)

## Linear mixed model fit by REML ['lmerMod']
## Formula: Telomeres_percent_change_C_to_D ~ Growth_rate_3_Months_Post +
## Treatment + (1 | Nest_ID)
## Data: subset_telo_cd
##
## REML criterion at convergence: 562.5
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -1.2958 -0.7633 -0.1910 0.4704 3.2203
##
## Random effects:
## Groups Name Variance Std.Dev.

```

```
## Nest_ID (Intercept)    0    0.00
## Residual              4803    69.31
## Number of obs: 52, groups: Nest_ID, 14
##
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept)      12.948    28.216   0.459
## Growth_rate_3_Months_Post -4.859    31.590  -0.154
## TreatmentConstant-Warmth  19.697    19.243   1.024
##
## Correlation of Fixed Effects:
##           (Intr) G__3_M
## Grwt__3_M_P -0.881
## TrtmntCns-W -0.304 -0.026
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
emmeans(model_telo_cd, pairwise ~ Treatment)
```

```
## $emmeans
## Treatment      emmean   SE   df lower.CL upper.CL
## Cold-Dormancy    9.09 13.4 32.7 -18.2871    36.5
## Constant-Warmth 28.78 14.1 30.9  0.0376    57.5
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast              estimate   SE   df t.ratio p.value
## (Cold-Dormancy) - (Constant-Warmth) -19.7 19.5 42.2 -1.009 0.3188
##
## Degrees-of-freedom method: kenward-roger
```

Data Analysis (Mitochondrial DNA density)

We will run three mixed-effect models at each interval with Percent change in mtDNA as a response variable and Treatment as a fixed effect, and Nest_ID as a random effect.

Hypothesis: Cold dormancy will suppress metabolic phenotypes including lower mitochondrial copy number and the suppressive effect will persist 3 weeks and 3 months after dormancy.

```
merged_data <- read.csv("../Data/GT_Growth_Telo_mtDNA.csv", na.strings = "NA")
str(merged_data)
```

```
## 'data.frame':   56 obs. of  13 variables:
## $ Nest_ID      : chr  "GT2023_N02" "GT2023_N02" "GT2023_N02" "GT2023_N03" ...
## $ Treatment    : chr  "Cold-Dormancy" "Cold-Dormancy" "Constant-Warmth" "Cold-Dormancy" ...
## $ Tortoise_ID  : chr  "GT2023_N02.02" "GT2023_N02.03" "GT2023_N02.04" "GT2023_N03.02" ...
## $ Tank         : int   3 11 5 6 1 8 9 10 11 1 ...
```

```
## $ Growth_rate_During      : num  0.0873 0.0232 0.7136 0.0687 0.1145 ...
## $ Growth_rate_3_Weeks_Post : num  0.547 0.626 1.037 0.789 0.479 ...
## $ Growth_rate_3_Months_Post : num  0.784 0.46 0.757 0.929 0.774 ...
## $ mtDNA_percent_change_A_to_B : num  NA 22.7 52.4 121.4 25.5 ...
## $ mtDNA_percent_change_B_to_C : num  -0.3 19.9 -2.22 -15.88 -31.53 ...
## $ mtDNA_percent_change_C_to_D : num  10.41 -47.21 -37.05 -43.76 -7.06 ...
## $ Telomeres_percent_change_A_to_B: num  NA 36.6 254.4 39.9 379 ...
## $ Telomeres_percent_change_B_to_C: num  -6.9 99.4 -48.6 64.6 -76.5 ...
## $ Telomeres_percent_change_C_to_D: num  117.07 -49.67 -43.96 -16.96 5.37 ...
```

```
##### Correlation for mtDNA in Winter (between Before and During) #####
```

```
# Subset to remove NAs for interval A to B
```

```
subset_mtDNA_ab <- subset(merged_data, !is.na(mtDNA_percent_change_A_to_B))
```

```
# Overall correlation
```

```
cor.test(subset_mtDNA_ab$mtDNA_percent_change_A_to_B, subset_mtDNA_ab$Growth_rate_During,
method = "spearman")
```

```
##
## Spearman's rank correlation rho
##
## data: subset_mtDNA_ab$mtDNA_percent_change_A_to_B and subset_mtDNA_ab$Growth_rate_During
## S = 7862, p-value = 0.1018
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0.2624765
```

```
# Correlation by treatment
```

```
by(subset_mtDNA_ab, subset_mtDNA_ab$Treatment, function(df) {
  cor.test(df$mtDNA_percent_change_A_to_B, df$Growth_rate_During, method = "spearman")
})
```

```
## subset_mtDNA_ab$Treatment: Cold-Dormancy
##
## Spearman's rank correlation rho
##
## data: df$mtDNA_percent_change_A_to_B and df$Growth_rate_During
## S = 1780, p-value = 0.4983
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## -0.1558442
## -----
## subset_mtDNA_ab$Treatment: Constant-Warmth
##
## Spearman's rank correlation rho
##
## data: df$mtDNA_percent_change_A_to_B and df$Growth_rate_During
## S = 998, p-value = 0.6104
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
```

```

##          rho
## 0.1245614

##### Correlation for mtDNA Early Spring (between During and 3 Weeks Post)
# Subset to remove NAs for interval B to C
subset_mtDNA_bc <- subset(merged_data, !is.na(mtDNA_percent_change_B_to_C))

# Overall correlation
cor.test(subset_mtDNA_bc$mtDNA_percent_change_B_to_C, subset_mtDNA_bc$Growth_rate_3_Weeks_Post,
method = "spearman")

##
## Spearman's rank correlation rho
##
## data: subset_mtDNA_bc$mtDNA_percent_change_B_to_C and subset_mtDNA_bc$Growth_rate_3_Weeks_Post
## S = 36041, p-value = 0.0006584
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##          rho
## -0.453028

# Correlation by treatment
by(subset_mtDNA_bc, subset_mtDNA_bc$Treatment, function(df) {
  cor.test(df$mtDNA_percent_change_B_to_C, df$Growth_rate_3_Weeks_Post, method = "spearman")
})

## subset_mtDNA_bc$Treatment: Cold-Dormancy
##
## Spearman's rank correlation rho
##
## data: df$mtDNA_percent_change_B_to_C and df$Growth_rate_3_Weeks_Post
## S = 3789.2, p-value = 0.4352
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##          rho
## -0.1566412
## -----
## subset_mtDNA_bc$Treatment: Constant-Warmth
##
## Spearman's rank correlation rho
##
## data: df$mtDNA_percent_change_B_to_C and df$Growth_rate_3_Weeks_Post
## S = 3881.2, p-value = 0.1031
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##          rho
## -0.3268935

##### Correlation for mtDNA Late Spring (Between 3 Weeks Post and 3 Months Post)
# Subset to remove NAs for interval C to D
subset_mtDNA_cd <- subset(merged_data, !is.na(mtDNA_percent_change_C_to_D))

```

```

# Overall correlation
cor.test(subset_mtDNA_cd$mtDNA_percent_change_C_to_D, subset_mtDNA_cd$Growth_rate_3_Months_Post,
method = "spearman")

##
## Spearman's rank correlation rho
##
## data: subset_mtDNA_cd$mtDNA_percent_change_C_to_D and subset_mtDNA_cd$Growth_rate_3_Months_Post
## S = 21592, p-value = 0.5812
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0.07829075

# Correlation by treatment
by(subset_mtDNA_cd, subset_mtDNA_cd$Treatment, function(df) {
  cor.test(df$mtDNA_percent_change_C_to_D, df$Growth_rate_3_Months_Post, method = "spearman")
})

## subset_mtDNA_cd$Treatment: Cold-Dormancy
##
## Spearman's rank correlation rho
##
## data: df$mtDNA_percent_change_C_to_D and df$Growth_rate_3_Months_Post
## S = 2834.9, p-value = 0.5031
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0.1346359
##
## -----
## subset_mtDNA_cd$Treatment: Constant-Warmth
##
## Spearman's rank correlation rho
##
## data: df$mtDNA_percent_change_C_to_D and df$Growth_rate_3_Months_Post
## S = 2496, p-value = 0.8497
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0.04

# Running a linear-regression for A to B interval
model_mtDNA_ab <- lmer(mtDNA_percent_change_A_to_B ~ Treatment + (1|Nest_ID),
data = subset_mtDNA_ab)
summary(model_mtDNA_ab)

## Linear mixed model fit by REML ['lmerMod']
## Formula: mtDNA_percent_change_A_to_B ~ Treatment + (1 | Nest_ID)
## Data: subset_mtDNA_ab
##

```

```
## REML criterion at convergence: 409.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.4045 -0.5194 -0.1880  0.2237  2.8938
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
## Nest_ID   (Intercept) 300.5    17.34
## Residual                2121.0   46.05
## Number of obs: 40, groups: Nest_ID, 14
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)         8.706     11.331  0.768
## TreatmentConstant-Warmth 23.886     14.950  1.598
##
## Correlation of Fixed Effects:
##              (Intr)
## TrtmntCns-W -0.633
```

```
emmeans(model_mtDNA_ab, pairwise ~ Treatment)
```

```
## $emmeans
## Treatment      emmean    SE    df lower.CL upper.CL
## Cold-Dormancy      8.71 11.6 23.5  -15.21    32.6
## Constant-Warmth    32.59 11.9 27.3    8.17    57.0
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast                                estimate    SE    df t.ratio p.value
## (Cold-Dormancy) - (Constant-Warmth)    -23.9 15.3 32.5  -1.561  0.1282
##
## Degrees-of-freedom method: kenward-roger
```

```
# Running a linear-regression for B to C interval
model_mtDNA_bc <- lmer(mtDNA_percent_change_B_to_C ~ Treatment + (1|Nest_ID),
data = subset_mtDNA_bc)
summary(model_mtDNA_bc)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: mtDNA_percent_change_B_to_C ~ Treatment + (1 | Nest_ID)
##      Data: subset_mtDNA_bc
##
## REML criterion at convergence: 491.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.32448 -0.67419 -0.08174  0.58364  2.15189
##
## Random effects:
```

```
## Groups Name Variance Std.Dev.
## Nest_ID (Intercept) 0.0 0.00
## Residual 790.9 28.12
## Number of obs: 53, groups: Nest_ID, 14
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 8.507 5.412 1.572
## TreatmentConstant-Warmth -27.141 7.727 -3.512
##
## Correlation of Fixed Effects:
## (Intr)
## TrtmntCns-W -0.700
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
emmeans(model_mtDNA_bc, pairwise ~ Treatment)
```

```
## $emmeans
## Treatment emmean SE df lower.CL upper.CL
## Cold-Dormancy 8.51 5.48 33.8 -2.63 19.65
## Constant-Warmth -18.63 5.63 30.2 -30.13 -7.14
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## (Cold-Dormancy) - (Constant-Warmth) 27.1 7.88 46.3 3.444 0.0012
##
## Degrees-of-freedom method: kenward-roger
```

```
# Running a linear-regression for C to D interval
model_mtDNA_cd <- lmer(mtDNA_percent_change_C_to_D ~ Treatment + (1|Nest_ID),
data = subset_mtDNA_cd)
summary(model_mtDNA_cd)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: mtDNA_percent_change_C_to_D ~ Treatment + (1 | Nest_ID)
## Data: subset_mtDNA_cd
##
## REML criterion at convergence: 490.7
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -1.5760 -0.9039 0.1263 0.8055 1.9710
##
## Random effects:
## Groups Name Variance Std.Dev.
## Nest_ID (Intercept) 0.0 0.00
## Residual 940.7 30.67
## Number of obs: 52, groups: Nest_ID, 14
##
```

```
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept)      -6.006      5.903  -1.018
## TreatmentConstant-Warmth  17.298      8.513   2.032
##
## Correlation of Fixed Effects:
##           (Intr)
## TrtmntCns-W -0.693
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
emmeans(model_mtDNA_cd, pairwise ~ Treatment)
```

```
## $emmeans
## Treatment      emmean   SE    df lower.CL upper.CL
## Cold-Dormancy   -6.01  5.95  34.2   -18.10     6.08
## Constant-Warmth  11.29  6.23  32.5    -1.39    23.97
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast              estimate    SE    df t.ratio p.value
## (Cold-Dormancy) - (Constant-Warmth)  -17.3  8.63  43.1  -2.005  0.0513
##
## Degrees-of-freedom method: kenward-roger
```

Plotting the data for Telomere length

```
plot_df <- bind_rows(
  subset_telo_ab %>%
    select(Treatment, Telomeres_percent_change_A_to_B) %>%
    mutate(Timepoint = "Winter", PercentChange = Telomeres_percent_change_A_to_B),

  subset_telo_bc %>%
    select(Treatment, Telomeres_percent_change_B_to_C) %>%
    mutate(Timepoint = "Early Spring", PercentChange = Telomeres_percent_change_B_to_C),

  subset_telo_cd %>%
    select(Treatment, Telomeres_percent_change_C_to_D) %>%
    mutate(Timepoint = "Late Spring", PercentChange = Telomeres_percent_change_C_to_D)
) %>%
  select(Treatment, Timepoint, PercentChange)

# Set timepoint order
plot_df$Timepoint <- factor(plot_df$Timepoint, levels = c("Winter", "Early Spring",
"Late Spring"))

# Summarize for plotting
summary_df <- plot_df %>%
  group_by(Treatment, Timepoint) %>%
```



```

summarise(
  mean = mean(PercentChange, na.rm = TRUE),
  se = sd(PercentChange, na.rm = TRUE) / sqrt(n()),
  .groups = "drop"
)

# Define color-blind friendly palette
cbbPalette <- c("#000000", "#E69F00", "#56B4E9", "#009E73", "#F0E442", "#0072B2",
"#D55E00", "#CC79A7")

# Define timepoint labels
timepoint_labels <- c(
  "Winter" = "Winter",
  "Early Spring" = "Early Spring",
  "Late Spring" = "Late Spring"
)

# Create the plot
telomeres_plot <- ggplot(plot_df, aes(x = Timepoint, y = PercentChange, color = Treatment)) +
  geom_boxplot(
    outlier.shape = NA,
    width = 0.6,
    position = position_dodge(0.8),
    fill = "white",
    alpha = 0.5,
    linewidth = 0.5
  ) +
  geom_jitter(
    position = position_jitterdodge(jitter.width = 0.1, dodge.width = 0.8),
    size = 2,
    alpha = 0.4
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 2,
    aes(group = Treatment),
    position = position_dodge(width = 0.8)
  ) +
  stat_summary(
    fun = mean,
    geom = "line",
    aes(group = Treatment),
    linewidth = 0.6,
    position = position_dodge(width = 0.8)
  ) +
  scale_color_manual(
    values = cbbPalette[c(6, 7)],
    name = "Treatment",
    labels = c("Cold-Dormancy", "Constant-Warmth")
  ) +
  scale_x_discrete(labels = timepoint_labels) +
  scale_y_continuous(breaks = seq(-100, 500, by = 50)) + # Adjust range if needed

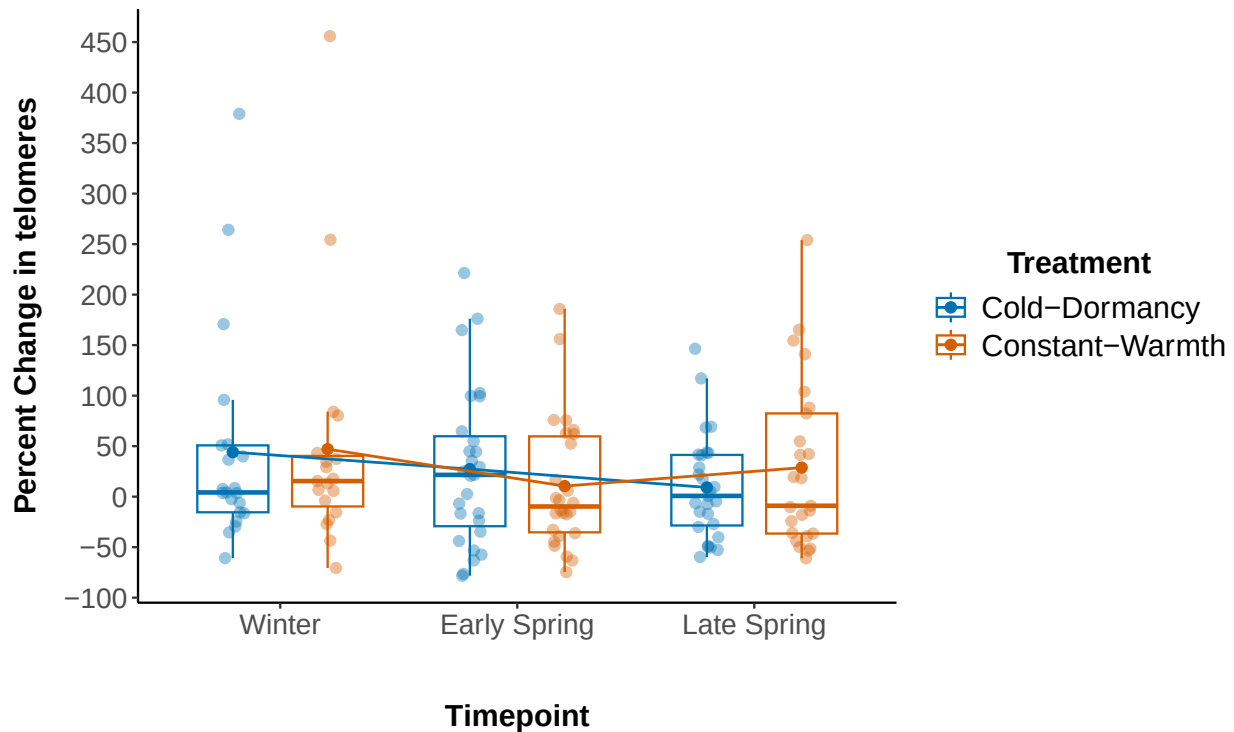
```

```

theme_classic() +
theme(
  panel.grid = element_blank(),
  axis.title.x = element_text(size = 14, face = "bold", margin = margin(t = 30, b = 15)),
  axis.title.y = element_text(size = 14, face = "bold", margin = margin(r = 10)),
  axis.text = element_text(size = 13),
  legend.title = element_text(size = 14, face = "bold"),
  legend.text = element_text(size = 14),
  legend.position = "right",
  legend.title.align = 0.5
) +
labs(
  x = "Timepoint",
  y = "Percent Change in telomeres"
)

```

telomeres_plot



```

# Save the plot as PNG for final figure production
ggsave(telomeres_plot, file = "./Figures/Telomere_length.png", width = 12, height = 6, dpi = 600)

```

Plotting the data for mtDNA density

```

plot_df <- bind_rows(
  subset_mtDNA_ab %>%

```

```

    select(Treatment, mtDNA_percent_change_A_to_B) %>%
    mutate(Timepoint = "Winter", PercentChange = mtDNA_percent_change_A_to_B),

subset_mtDNA_bc %>%
    select(Treatment, mtDNA_percent_change_B_to_C) %>%
    mutate(Timepoint = "Early Spring", PercentChange = mtDNA_percent_change_B_to_C),

subset_mtDNA_cd %>%
    select(Treatment, mtDNA_percent_change_C_to_D) %>%
    mutate(Timepoint = "Late Spring", PercentChange = mtDNA_percent_change_C_to_D)
) %>%
select(Treatment, Timepoint, PercentChange)

# Set timepoint order
plot_df$Timepoint <- factor(plot_df$Timepoint, levels = c("Winter", "Early Spring",
"Late Spring"))

# Summarize for plotting
summary_df <- plot_df %>%
  group_by(Treatment, Timepoint) %>%
  summarise(
    mean = mean(PercentChange, na.rm = TRUE),
    se = sd(PercentChange, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

# Define color-blind friendly palette
cbbPalette <- c("#000000", "#E69F00", "#56B4E9", "#009E73", "#F0E442", "#0072B2",
"#D55E00", "#CC79A7")

# Define timepoint labels
timepoint_labels <- c(
  "Winter" = "Winter",
  "Early Spring" = "Early Spring",
  "Late Spring" = "Late Spring"
)

# Create the plot
mtDNA_plot <- ggplot(plot_df, aes(x = Timepoint, y = PercentChange, color = Treatment)) +
  geom_boxplot(
    outlier.shape = NA,
    width = 0.6,
    position = position_dodge(0.8),
    fill = "white",
    alpha = 0.5,
    linewidth = 0.5
  ) +
  geom_jitter(
    position = position_jitterdodge(jitter.width = 0.1, dodge.width = 0.8),
    size = 2,
    alpha = 0.4
  ) +

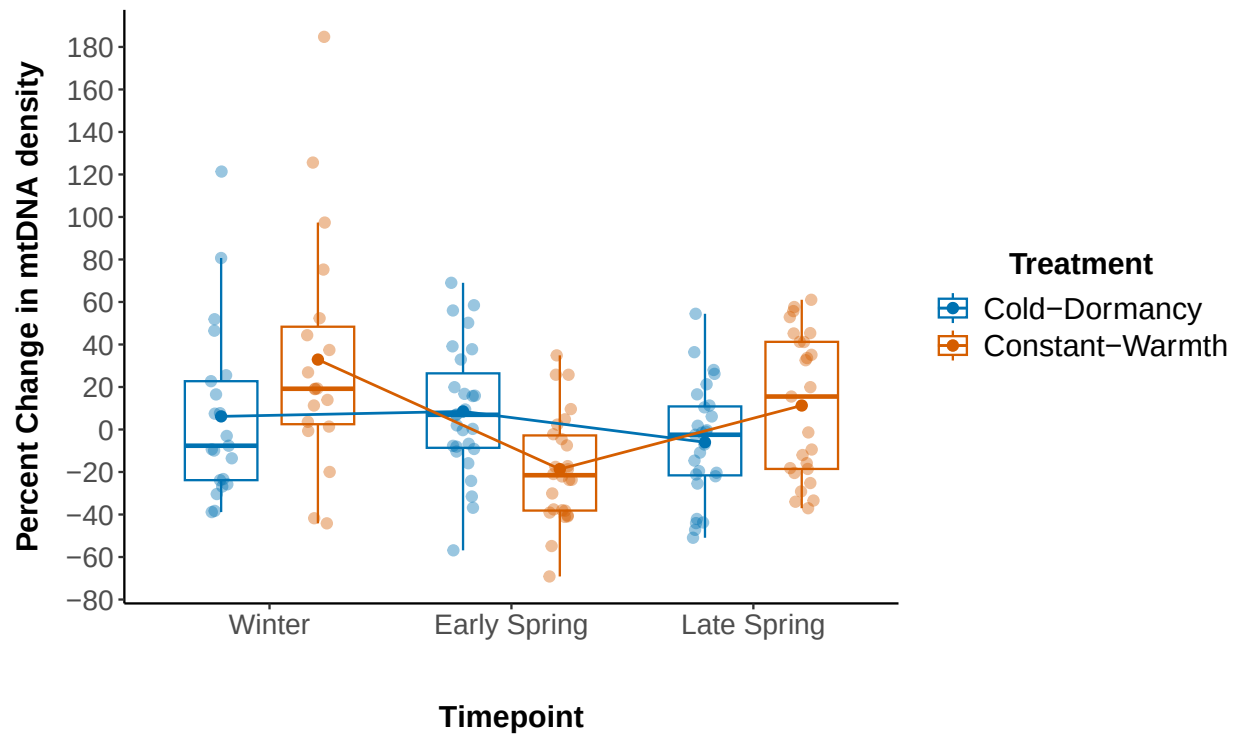
```

```

stat_summary(
  fun = mean,
  geom = "point",
  size = 2,
  aes(group = Treatment),
  position = position_dodge(width = 0.8)
) +
stat_summary(
  fun = mean,
  geom = "line",
  aes(group = Treatment),
  linewidth = 0.6,
  position = position_dodge(width = 0.8)
) +
scale_color_manual(
  values = cbbPalette[c(6, 7)],
  name = "Treatment",
  labels = c("Cold-Dormancy", "Constant-Warmth")
) +
scale_x_discrete(labels = timepoint_labels) +
scale_y_continuous(breaks = seq(-100, 200, by = 20)) + # Adjust range if needed
theme_classic() +
theme(
  panel.grid = element_blank(),
  axis.title.x = element_text(size = 14, face = "bold", margin = margin(t = 30, b = 15)),
  axis.title.y = element_text(size = 14, face = "bold", margin = margin(r = 10)),
  axis.text = element_text(size = 13),
  legend.title = element_text(size = 14, face = "bold"),
  legend.text = element_text(size = 14),
  legend.position = "right",
  legend.title.align = 0.5
) +
labs(
  x = "Timepoint",
  y = "Percent Change in mtDNA density"
)

mtDNA_plot

```



```
# Save the plot as PNG for final figure production
ggsave(mtDNA_plot, file = "./Figures/mtDNA_density.png", width = 12, height = 6, dpi = 600)
```

This is a link to the GitHub repository

Gopher_tortoise_Head-start_2023